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exploring the diversity of consciousness

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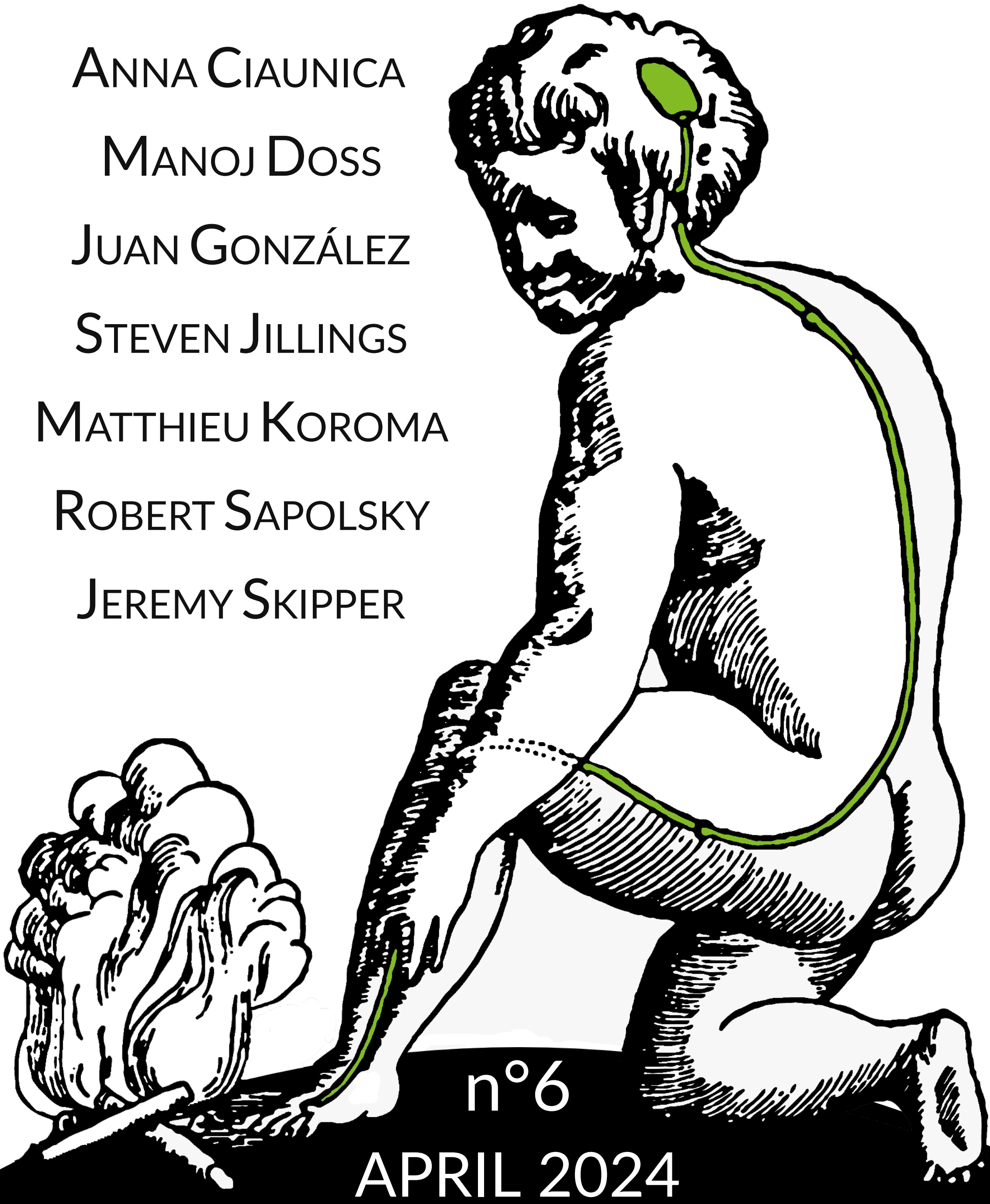

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# What is it like to be in early sensory life?

## An episode with Anna Ciaunica

Recorded by Daniel Friedman and edited by Lena Coutrot and Matthieu Koroma

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## Abstract

In this episode, Anna Ciaunica reviews the recent progress on the scientific study of conscious experiences in early sensory life.

***keywords***: *Early Sensory Life, Consciousness, Multisensory, Embodiment, Shared Experience*

The video and audio version of this episode are available on https://aliusresearch.org/bulletin06-ciaunica-podcast.html

Consciousness is a mystery that many philosophers and scientists have attempted to tackle throughout the centuries. But what if instead of asking what consciousness *is*, one asks how conscious experiences *emerge*, *develop* and *unfold* in our life journey? Let's go back to 'square one' and look at how experiences arise in early human life. What is it like to experience and perceive the world at the dawn of our lives? One basic yet overlooked aspect of current discussions in both philosophy and cognitive neuroscience is that experiences first develop *within* another human body. The most basic perceptions and actions emerge already *in utero*. Crucially, while not all humans will have the experience of being pregnant or carrying a baby, the experience of *perceiving and growing within another person's body* is universal. Why is this important? How can this observation help us to understand conscious experiences?

> " One basic yet overlooked aspect of current discussions in both philosophy and cognitive neuroscience is that experiences first develop *within* another human body. "

When I look outside my window, as an adult, I can see trees, hear the noise of the cars, smell the rain. All these perceptions form the content of my conscious experiences. But in the womb, starting from 22 weeks onwards, the world looks different. How? First, my entire body is immersed in a liquid environment, which makes my movements literally fluid. My eyes are mostly closed, and even if the light comes in through my closed eyelids, my visual perception is limited. The proximal senses, such as touch and smell, are taking the lead in exploring the world. I can hear things, but most importantly, I can hear someone's heart beating all the time, continuously, rhythmically. The early sensory world of a human is not isolated nor solitary. It is a rich multisensory experiential orchestra with several players, involving others' embodied presence, experiences and moving bodies. Perceiving the world is not just about me, one individual looking at the world through a window. At the beginning of our lives, perceiving is about co-perceiving a shared world. We literally share time and space and our bodies with another human being.

“ The early sensory world (…) is a rich multisensory experiential orchestra with several players, involving others' embodied presence, experiences and moving bodies. ”

This observation is in line with recent work in mind and brain research stipulating that our perceptions, cognitions and actions are fundamentally geared towards biological self-preservation, *i.e.*, the need to maintain and regulate the psychological and physiological needs for the integrity of the living organism (the human body or individual) within a wider and highly volatile social and physical environment (Varela et al. 1991). This is something we share with all living organisms. In order to fulfil the fundamental conditions for self-preservation, humans need to constantly move, act and interact with the physical and social environment. Now, one way the human organism may complete this task is described by the influential 'Predictive Processing' framework (Friston 2010; Clark 2013; Hohwy 2013). The idea here is that we do not perceive the world as it is. Rather we perceive it as we 'expect' or 'predict' it will be, based on prior perceptions and experiences.

If this is so, then in order to understand the nature of conscious experience in the here and now of adult life, it is essential to look at how these experiences get off the ground from the outset, in early life (Ciaunica et al. 2021a, 2021b). This is because adult conscious experiences cannot be addressed in isolation of prior (early life) perceptual experiences. Regardless of when various forms of consciousness first emerge, there is an experiential continuum between early and later experiences. The key idea here is that by endorsing a bottom-up, sensory and developmental perspective in exploring how conscious experiences dynamically arise and develop in concert with the developing organism, we may gain important insights into what consciousness 'is' and its biological structure. Unveiling the mystery of consciousness perhaps then will depend less on what makes human experiences uniquely special, but rather on what connects them with life under all its forms.

> " Adult conscious experiences cannot be addressed in isolation of prior (early life) perceptual experiences. "

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# On The Cinco Sins of Psychedelic Research

## An interview with Manoj Doss

By George Fejer

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### Abstract

In this interview, Manoj Doss addresses the "Cinco Sins of Psychedelic Research," highlighting methodological concerns and biases, including those stemming from self-experimentation and researchers' personal drug experiences. Advocating for a closer integration of cognitive psychology and neuropharmacology, he calls for empirical approaches to study psychedelics' effects on cognition and memory. Doss's critique underscores the importance of behavioral measures to advance a scientific understanding of psychedelic substances.

***Keywords***: *Cognitive Neuropsychopharmacology, Psychedelic Research, Bias, Subjectivity*

Hello Manoj, thank you for taking the time to do this interview. I wanted to use this opportunity to ask you some questions, given your reputation for having a critical voice within the psychedelic research community. You have raised some sharp concerns over issues such as personal biases and methodological rigor, as outlined in your lectures on the "Cinco Sins of Psychedelics" (e.g., Morris, 2022). In your line of research, you have investigated the effects of psychedelics and various psychoactive drugs on episodic memory, among many other interesting topics. But I wanted to open up the conversation by discussing your job title, which you have ascribed as being a "Cognitive Neuropsychopharmacologist". What does this entail?

Basically, I have been using that title as a joke to give myself the longest career title possible. Some researchers call themselves Cognitive Neuroscientists, others call themselves Neuropsychopharmacologists, but I identify as both since I am a cognitive psychologist who measures the brain while under the influence of different drugs.

Can you give an example of the type of research questions that motivate you in trying to bring these two disciplines closer together? Are you more interested in characterizing the effects of drugs by looking at how they affect memory, or in probing the underlying processes of memory through drug interventions? What started your initial interest in entering the field?

Most pharmacologists tend to stay clear of cognitive psychology, yet I believe this field holds significant value, particularly when examining drugs with bizarre mental effects by investigating how they affect information processing. What started my initial interest was the investigation of episodic memory, a subdomain of cognitive psychology that has been studied for over a century. It is possibly one of the oldest research topics in the field, and as such, it has developed stringent methods. But cognitive psychologists typically avoid psychopharmacology, because drug interventions are considered as a very messy type of whole-brain manipulation. However, in my opinion, this is an excellent place to examine pitfalls in the field of psychopharmacology, as I have discovered that many researchers are not well-trained in measuring cognitive processes like memory. It can be difficult to be good at everything, but I believe that cognitive psychology offers a lot of valuable tools. On the flip side, the field of cognitive psychology can often be pedantic, with researchers debating the smallest effect sizes which do not necessarily have any immediate implications for improving something in the world. Pursuing a career in psychopharmacology next to cognitive psychology was a good way for me to broaden my research interests and make them more relevant.

> “One of the other major benefits of cognitive psychology […] is that you can try to set yourself up to fail. […] In contrast, psychedelic drug studies … investigating subjective effects do not take the risk of producing null results.”

To date ,the most high-impact paper that I have published was also the simplest data set to analyze and collect. It demonstrated that when people retrieve memories under the influence of THC, they are more likely to have a false memory due to an increased false alarm rate (Doss et al., 2018). There was no fancy analysis or brain imaging in this paper, but it was accepted for publication in the journal *Biological Psychiatry* because no one in the field had ever properly isolated the drug effects in relation to the retrieval phase of a memory test. I believe there is a lot of such low-hanging fruit, where simple behavioral paradigms can go a long way in explaining how drugs affect behavior and cognition. That may sound opportunistic, but I am in fact very interested in differentiating the effects of different drugs using precise cognitive manipulations.

> “ [Y]ou cannot really use drugs a lot of times to show how the mind works. I was trying to prove him wrong at first, but now I am beginning to believe he is correct. ”

One of the other major benefits of cognitive psychology, in my opinion, is that you can try to set yourself up to fail. A researcher can run an experiment without necessarily expecting to find any effect, but if they find an effect nonetheless, that's probably a good sign. In contrast, many psychedelic drug studies set themselves up for success because they are all about investigating subjective effects and do not take the risk of producing null results. People report a wide range of subjective effects, so the researchers create a questionnaire to assess things like mystical experiences, administer the drug alongside the questionnaire, and discover that people score these higher while under the influence of the drug. What else would they anticipate was happening? Did they expect to find any negative results? If you give people a lot of different questionnaires while under the influence of drugs, they will probably score higher on at least some of them. To me, it would seem much more interesting to read a study that produced unexpected results, and using methods of cognitive psychology is a good way to do that.

### Is there something particularly insightful about using drug interventions in the field of cognitive psychology, or more specifically, memory research?

Charan Ranganath, my advisor at UC Davis, once told me that you cannot really use drugs a lot of times to show how the mind works (although he does not remember saying that). I was trying to prove him wrong at first, but now I am beginning to believe he is correct. I have not learned much about the mind from studying drugs, but I am still trying. There are numerous confirmatory findings of things we already knew but did not need drug manipulation to prove. It is great that data gathered through drug manipulations converge on those conclusions. Take, for example, one of my recent publications (Doss et al., 2023), which showed that psychedelics can selectively impair memory recollection but spare (or enhance) familiarity even at high doses, while other drugs impaired both processes. This suggests these memory functions are two distinct processes, but there were already lesion populations showing selective impairments of each memory process and loads of fMRI and EEG research that found evidence for their dissociation. But imagine if we only had evidence for this dissociation from drug findings. It would make you wonder if such findings can even be applied to understanding a typical brain. Nonetheless, I do not think pharmacology will ultimately teach me anything new about memory. Of course, I could be wrong, and I am still open to that possibility.

Another thing I am interested in is the impact of drugs on memory reconsolidation, a process where retrieved memories become destabilized and altered before being re-stored in a modified state, particularly due to the clinical implications of memory alteration. Memory reconsolidation allows for the updating and integration of new information into existing memories. Research has shown that when people retrieve memories, their mental state becomes very susceptible to external manipulation, such that it is possible to delete specific memories (Haubrich & Nader, 2018). But it is very difficult to trigger memory reconsolidation processes in humans and much easier in lab rats. One reason for this might be that lab rats have a very homogeneous life, and do not have a lot of other memories. Any type of stimulus used for the induction of new memories, for instance, a loud buzzing sound associated with an electric shock, is likely to be the first time it encounters that sound. I also wonder whether it would be different in a rat population from the New York subway where they are quite used to such sounds. But in lab rats who are not used to such stimuli, it is very simple to trigger the reconsolidation of fear by re-administering the same sound. By contrast, humans have vastly more associations with any type of stimulus, which makes it very difficult to trigger memory reconsolidation processes in a targeted manner. I would be interested in whether psychedelics (or any other type of drugs) could strengthen or destabilize memories in some manner that makes memory reconsolidation easier.

In your critique of research on the subjective effects of psychedelics, you suggest that many researchers are primarily motivated to confirm their own positive experiences with drugs are true. You have criticized the belief in the psychedelic research field that these drugs reveal new insights into brain or mind structure, stating that research has mainly shown how these substances affect the mind without discovering any new underlying principles (Love, 2019. You also noted that scientists privately admit to using psychedelics but avoid public disclosure due to legal issues and fear of losing funding. This seems to paint a picture, that the research is biased  because researchers do not disclose their personal experiences with drug use. However, looking at it from another angle, those researchers might argue that their personal experiences can improve the quality of their work. Even though they have personal experiences, their research findings are still based on empirical evidence and do not only reflect their own experiences, even if that was the motivation to research it. This situation might be similar to researchers who decide to study mental health issues like depression because they or their family members have gone through it. It would be interesting to know what drove you to research psychedelics in the first place.

When it comes to drugs in general, I had some curiosity since high school. I used to hang around skater kids and participate in rollerblading doing tricks and flips. It was a countercultural scene, and I recall encountering weed around seventh or eighth grade. Initially, I strongly opposed it and believed that skating was all I needed. However, I noticed friends quitting skating for drugs, which made me question why people chose that path. It was interesting yet concerning to see the negative consequences some of my friends experienced. The first time I seriously took notice of hallucinogens was when a friend from high school took dextromethorphan and later tragically took his own life. This incident made me reflect on the bizarre effects of hallucinogens, such as delusional thinking. Although I was initially against drugs, I found myself strangely fascinated by them, similar to a morbid fear and fascination towards snakes. Over time, my perspective changed, and I realized that people should have the freedom to make their choices as long as they don't harm others. I also recognize that every class of drugs serves a medical purpose, including ketamine, dextromethorphan, and opioids. While some drugs are safer than others, psychedelics seem to be the exception at the moment, but I think it is essential to explore and compare different psychoactive drugs to gain a comprehensive understanding of their unique effects and potential therapeutic uses.

Thank you for sharing this history. Following up on my previous question, could you elaborate on why you think first-hand experiences lead to biases and bad science? Could you give some examples?

I agree that for some researchers, their initial experience with drugs is what led them down the path of studying them, and this is largely independent of whether or not it taught them anything about how the brain, mind, or even the drugs themselves function. I can relate this to a concussion I had from a skating accident, which did not teach me much about memory, but the experience got me interested in studying it. The main problem is perhaps that we really cannot discuss our personal experiences openly. But on the other hand, I still have the impression that a lot of people enter the field primarily because of their past drug use, and not because they have a deeper scientific interest in neuroscience or psychology. This is similar to how many people want to become neuroscientists because putting pretty brain images makes the science more convincing (McCabe & Castel, 2008).

> “[M]any studies seem biased, because they are only designed to confirm predictable outcomes. They create the illusion of progress, but I am not convinced we are learning anything new by confirming that psychedelics are *psychedelic*.”

My critique of the psychedelic field is largely due to how the nature of the work creates the impression of bias. I hesitate to label it as 'bad science'. It is not intended as a criticism of individual researchers, but rather the approach the psychedelic field takes as a whole. In this respect, I think some biases affect researchers' perspectives and lead to less insightful study designs. As I have noted before, many studies seem biased, because they are only designed to confirm predictable outcomes. They create the illusion of progress, but I am not convinced we are learning anything new by confirming that psychedelics are *psychedelic*. There are various examples of this in recent research.

An obvious example is a recent study that reported improvements in racial trauma and mental health symptoms following psychedelic experiences (Williams et al., 2021) and they recruited only participants who attributed improvements in racial trauma to psychedelics, already before data collection. This would be very similar to conducting a study where we recruit people to treat their depression with psilocybin, and then conclude that psilocybin-seeking people have depression. It is a circular form of reasoning to report that psychedelics reduce racial trauma, in people who report improvements in racial trauma after taking psychedelics, and I think the primary motivation of this type of research is to prove something that the researchers already believe is true.

Another less obvious example is research that shows that LSD makes people react more emotionally in response to music (Kaelen et al., 2015). These results seem just as likely replicated using other psychoactive substances. Just name a recreational drug under the sun that is not used at music concerts. Someone recently told me they take Adderall at concerts, and this caught me off guard, even though it later occurred to me this is probably not that much different from taking cocaine at a music event. I think there is a clear design flaw in claiming that LSD makes music more emotional while not comparing other substances. A balanced approach may be to play different types of music under the influence of different substances. If this type of research showed us the result, for instance, that people started liking electronic music after taking psychedelics, that would be a much more remarkable effect.

Finally, I also bring up the study that showed that microdosing LSD dilates the perception of time perception (Yanakieva et al., 2019). The study had participants reproduce the duration of a stimulus (a blue circle) after viewing it. Under LSD, participants tended to reproduce longer intervals. However, this does not prove that psychedelics dilate time perception. If time perception were truly dilated, both the viewing and reproduction phases would be affected equally. This situation is akin to the El Greco fallacy, where it was wrongly assumed that El Greco's elongated painting style was due to a vision problem, not considering that this would have also altered his perception of the canvas. Interestingly, under LSD, they found that participants' estimates of time duration became more accurate, suggesting an improved alignment of perceived time with memory. The study's design conflated the encoding and retrieval phases of memory, making it unclear whether LSD affects the encoding of time duration or the retrieval process. A study of how these phases are affected separately might be more insightful. Additionally, earlier research (Wackermann et al., 2008) indicated that high and low doses of psilocybin might have the opposite effect on time perception, which raises questions about whether the emphasis on encoding vs. reproduction might shift under different psychedelic doses. Going back to my main point though, I think this case demonstrates how the design and implementation of a study would benefit from expertise in cognitive psychology.

Overall, I think such biases should be acknowledged, and we all need to be more critical of each other. I remain skeptical, as I have yet to hear someone explain how their drug experiences have provided insights into the workings of the mind or how drugs work.

> “ I think personal experiences may be helpful in examining some basic low-level perceptual phenomena such as color constancy. A researcher skilled in perceptual studies ... might gain valuable insights from a psychedelic experience. ”

Could we entertain the opposite idea, the possibility that having personal experience with psychedelics might make research better? Do you think there are situations where a researcher's own experiences with these drugs could help shape their research in a way that leads to important discoveries or ideas?

I think personal experiences may be helpful in examining some basic low-level perceptual phenomena such as color constancy. A researcher skilled in perceptual studies, particularly knowledgeable about the complexities of the magnocellular or parvocellular pathways, might gain valuable insights from a psychedelic experience. Descriptions of psychedelic experiences often suggest they disrupt color constancy, yet the narrators lack the specific vocabulary to pinpoint it. When I initially discovered this concept in psychophysics, I recognized that it might clarify certain psychedelic effects.

An expert in visual perception and color constancy could potentially discern the underlying cause of these alterations. Should they experiment with psychedelic substances without any preconceptions, they might question: why does a wall appear purple? Why do violet tones seem more prominent than greens and reds? Understanding that a white wall reflects a spectrum of colors, they might ponder why psychedelics accentuate the violets. Typically, the perception of specific colors hinges on external factors like the lighting conditions or surrounding colors. Such a researcher might examine these processes and inquire if these are typical characteristics of psychedelic experiences, or if they vary based on the context. For example, if an individual experiences a heightened perception of a wall's violet hues under LSD at a certain time, would the same hues be as pronounced under different circumstances?

This field remains largely unexplored, thus a deep comprehension of visual perception combined with a psychedelic experience might offer critical insights for research. Therefore, investigating how psychedelics disrupt certain aspects like constancy, detection of peripheral motion, or other perceptual distortions may be a worthwhile effort. It is conceivable that an individual with extensive knowledge of perception could formulate a theory for the drug's action post-experience. However, the bizarre, extraordinary, or unexplained phenomena reported by users draw more attention.

For many people, the value of their psychedelic experiences lies more in the deep insights they gain about their views on life, rather than just changes in basic perception or how they process information. How do you view the scientific efforts to explore these deeper, "high-level" experiences?

For example, your colleague David Yaden speaks of the noetic qualities of psychedelic experiences (Yaden et al., 2021; Yaden et al., 2017). Fortier-Davy and Millière (2020) also note that the heightened 'sense of reality' which is characteristic of psychedelics, is an important factor in distinguishing them from hallucinations induced through anticholinergics, or hallucinations in psychosis-like states. Psychedelic hallucinations seem to be much more lucid compared to anti-cholinergic hallucinations (Fortier, 2018), where people become completely detached from reality and lack basic insights into the fact that they are even hallucinating. What do you think about the study of these phenomena, the current frameworks that investigate psychedelic drug actions in terms of alterations in higher-level, or top-down, aspects of cognition?

First of all, people may just as well encounter delusional states when under the influence of psychedelics, such as experiencing complete disconnection from reality with high doses of psilocybin or vivid hallucinations of entities like machine elves with inhaled DMT. Even at lower doses, individuals can develop delusional beliefs or a misleading sense of insight, believing they have gained a profound understanding of the universe or consciousness that often doesn't translate into their everyday lives. People may report that things feel more real, more important, or more relevant to them, but I maintain that it is crucial to approach these claims with skepticism, akin to how one would view similar reports from individuals with schizophrenia, recognizing that a heightened sense of reality does not equate to actual reality. But for some reason when people report seeing machine elves on DMT, some researchers seem to take the notion that they may exist, seriously.

> “I maintain that it is crucial to approach these claims with skepticism, akin to … reports from individuals with schizophrenia, recognizing that a heightened sense of reality does not equate to actual reality. But for some reason when people report seeing machine elves on DMT, some researchers seem to take the notion that they may exist, seriously.”

Regarding your main question, I take a lot of inspiration from Endel Tulving (1985) who introduced the term "noetic consciousness" within the context of memory research. His work also introduced the distinction between episodic and semantic memory, closely linking episodic memory with the concept of autonoetic consciousness and semantic memory with noetic consciousness. Episodic memory allows individuals to mentally travel in time, recalling past experiences or imagining future ones, a process heavily reliant on the hippocampus, and involving a form of consciousness that entails an intact autobiographical self. On the other hand, noetic consciousness, associated with semantic memory, involves a basic sense of knowing or familiarity that operates independently of the hippocampus, enabling recognition without personal experiential recall. For instance, if we show people a list of words, and then disrupt the hippocampus they would still have a vague sense of which items were more or less familiar. If you take the hippocampus offline, then this sense of familiarity can be attached to irrelevant information. This is also in line with research demonstrating how the feeling of insight can be increased via semantic priming manipulation and misattributed to irrelevant ideas (Grimmer et al., 2022; McGovern et al., 2023). Thus it seems worth exploring how psychedelics operate, either by taking the hippocampus offline or through some other mechanism that boosts the sense of familiarity to test whether they create false insights or memories. The notion that psychedelic experiences provide a heightened sense of reality or true insight is, in my view, questionable, and we should just as well investigate whether they cause misattribution that can then lead to a belief in the reality or importance of these ideas or insights, despite a lack of objective evidence to support them, which is rather delusional.

In my research, I am most interested in uncovering new and fascinating aspects of these drugs, for instance, if they can enhance or impair memory, whether they induce sleep or wakefulness, and a variety of other effects. When investigating a drug that produces very strong subjective effects, I think it crucial to compare them with other psychoactive drugs that have similar characteristics. For example, psychedelics increase wakefulness, so why are we not comparing their effects with stimulants? Psychedelics also cause hallucinations, so we could also compare them with other NMDA antagonists like ketamine. But grouping ketamine, and serotonergic hallucinogens in one group as "psychedelics" based on similarities in their subjective effects does not advance our understanding of their effects.

**What is your take on more ambitious frameworks like the relaxed beliefs under psychedelics (REBUS) model (Carhart-Harris & Friston, 2019) that try to explain higher-order phenomena within the framework of predictive processing (Wiese & Metzinger, 2017) or the Free Energy Principle (Friston, 2010)? Do these frameworks provide a solid scientific foundation for exploring how subjective experiences, like the sensation of insight, correspond to mechanisms, such as the reduction of prediction error?**

I think the REBUS model shares the same shortcomings as predictive processing, which remains *unfalsifiable* (Friston et al., 2018). It seems like loads of subjective phenomena are thrown onto a prediction error: Oh, it's! It's rewarding! Oh, it produces insight! Oh, it produces surprise. Oh, it produces fear. Just make up your goddamn mind! Speaking to the notion that feelings of insight could be a result of specific mechanistic processes such as elevated rates of prediction error, or priors being updated at an uncommonly high level, the answer is not clear-cut. Various circuits form some kind of prediction error but I don't know how you can map something like that onto various subjective effects, certainly not a single one. I am sure it varies quite a bit by circuit. I am generally skeptical regarding its applicability due to this over-attribution of various subjective phenomena to prediction error.

My other critical issue with the REBUS model is its apparent lack of functional alignment and precision necessary to accurately map and predict the effects of psychedelics on cognition and behavior. The model puts forth the idea that under the influence of psychedelics, high-level cognition should be more impaired than low-level cognition. Yet, psychedelics can disrupt low-level learning forms, such as prepulse inhibition, while potentially enhancing higher-level functions like mnemonic familiarity (Doss et al., 2022; Doss et al., 2023). The REBUS model's claim that the hippocampus is a low-level structure, and the connectivity between the hippocampus and the default mode network increases under psychedelics, does not fully align with observed data. Studies show that impairment of the perirhinal cortex but not the hippocampus leads to impairments of familiarity (Bowles et al., 2007). Consequently, if psychedelics enhance familiarity, it suggests the engagement of high-level processes, not low-level ones as suggested by the model.

“ The way REBUS is written seems to be agnostic to cognitive psychology in many ways, except when it suits its purposes. ”

REBUS, much like the Free Energy Principle, tries to be a theory of everything. REBUS is supposed to be an explanation at the computational level. Until you can show me some very specific behaviors, I don't think that I'm going to buy into it. While the REBUS model's concept of heightened bottom-up prediction error aligns with reports from individuals under the influence of psychedelics — who often describe everything as being novel or surprising — it falls short in explaining other psychedelic experiences, such as déjà vu. Some of the things you'd expect from REBUS like greater brain connectivity and greater brain states are found in patients with obsessive-compulsive disorder (OCD). Are they always tripping out then? Even if you take the brain measures themselves, such as entropy, they don't have any specificity. Entropy increases with coffee. There's not a one-to-one mapping between the subjective experience, but it's meant to be somewhat expansive. The framework starts to break down when you begin attributing real functions to these brain predictions. The way REBUS is written seems to be agnostic to cognitive psychology in many ways, except when it suits its purposes. Currently, there's a tendency to generalize findings based on a small number of participants. Everybody's piggybacking on REBUS instead of challenging it. I think that's what we should be doing - challenging different theories.

Before the REBUS model was introduced, Vollenweider and Geyer (2001) had already proposed the idea that psychedelics could disrupt thalamic gating, causing a shift from top-down to bottom-up cognition. Nicholas Langlitz (2007) traces this idea's lineage back to Aldous Huxley's perennial concept of the mind as a "reducing valve" that filters sensory information to create a coherent reality. Huxley suggested that psychedelics expand this valve, allowing for a broader spectrum of perception and consciousness by permitting a more unfiltered influx of sensory data. Considering this backdrop and the hypothesis that psychedelics flatten or disrupt high-level priors, potentially enhancing susceptibility to priming and altering the content of hallucinations, what are your broader thoughts on this framework? How do you view the interplay between top-down and bottom-up processes in this context, and do you believe it offers a valid avenue for experimental exploration? How might this altered susceptibility to priming and the selective disruption of higher-level information affect the nature of hallucinations and individuals' susceptibility to them?

What I do not like about the notion of the reducing valve is that it could be interpreted as seeing things that are there, without distortions in your visual field, and the notion that everything is due to a relaxation of priors, like visual distortions. In theory, certain thoughts may be more likely to occur that would normally be filtered out. It would be interesting to explore this hypothesis by comparing hallucinations across cultures, for instance, whether people from Brazil are more likely to hallucinate jaguars on Ayahuasca compared with someone from a Western background. But on the other hand, such hallucinations require high-level representations which are allegedly disrupted by psychedelics. Despite predictions of disrupted high-level representations, such hallucinations continue to appear even on really high doses. People report visions of complex landscapes or encounters with deceased relatives, indicating that high-level concepts remain intact under psychedelics, which challenges the "flattened landscape" metaphor. This raises the question of where the hierarchy begins and ends, of what is considered low-level versus high-level. For instance, on a behavioral level, psilocybin impairs the perception of motion (Carter et al., 2004) whereas alcohol does the opposite (Wang et al., 2018), but in my view, these are still pretty low levels of information processing. Then there are instances of certain high-level functions being facilitated, like semantic priming (Spitzer et al., 1996), which is further upstream than these visual effects. Therefore, I propose that there are likely cases of facilitated top-down processing under psychedelics, which also do not comply with the reducing valve or thalamic gating model.

Looking at it on the circuit level, there is a hierarchy of back-and-forth projections throughout the cortex, and as you move from lower to higher levels of the cortex, there are also corresponding levels in the thalamus receiving and projecting back. The expected impairment of higher-order cognition being more pronounced than lower-order cognition is contradicted by the fact that even basic forms of learning, such as prepulse inhibition, are disrupted by psychedelics. From a subjective standpoint, I would also argue that mnemonic processes such as familiarity are enhanced by psychedelic substances and that these depend on structures such as the perirhinal cortex which is situated at the top of the ventral visual stream. This does not align well with the idea that psychedelics break down or flatten a prediction hierarchy. One of the explicit predictions from the REBUS model is that there should be greater effective connectivity, in other words, greater information flow from the hippocampus to the cortex. But this doesn't make sense considering hippocampus-dependent memory or recollection is impaired by psychedelics. Familiarity or the feeling of knowing is more dependent on the cortex. We tried to make sense of the REBUS model in terms of the circuit implementation. When it comes to the circuits, I think the cortico-striatal thalamo-cortical (CSTC) model (Doss et al., 2021) has the most evidence behind it and it does have some explanatory value for certain phenomena, like impairments in attention and inability to focus on something.

> “In discussions with Phil Corlett, I have contemplated whether the neural dynamics under psychedelics resemble more of a "bumpy seesaw" than a "flattened landscape."”

In discussions with Phil Corlett, I have contemplated whether the neural dynamics under psychedelics resemble more of a "bumpy seesaw" than a "flattened landscape." This "bumpy seesaw" metaphor, considering the slope as the flow of information within the hierarchy, might more accurately reflect the interplay between top-down and bottom-up information processing. While a flattened landscape suggests a uniform distribution of influence, the seesaw model implies a dynamic balance, with information oscillating between higher and lower processing levels. This back-and-forth motion suggests that both bottom-up and top-down information can penetrate more readily under the influence of psychedelics, challenging the notion of a completely flattened hierarchy. Essentially, this interaction includes a complex network of projections within the cortex and between the cortex and the thalamus, allowing for a nuanced exchange of information across different levels of cognition.

### Can you outline your main critique of studies aiming to establish a link between the subjective effects of drugs and resting-state brain connectivity?

Numerous studies have attempted to establish a link between the subjective effects of drugs and brain activity during periods of rest, but a key issue is the lack of replication in the observed correlations. For instance, while one study by Carhart-Harris et al. (2016) found that decreases in default mode network connectivity correlate with ego dissolution, another study by Müller et al. (2018) failed to replicate this finding despite controlling for head motion and having more subjects. This lack of consistency highlights the challenge of establishing a one-to-one mapping between brain activity and subjective drug effects.

My takeaway from this, is researchers must constrain their search space and exert more experimental control to gain a better understanding of the effects of drugs on the mind and brain. Although differences in retrospective reports, behavior, and brain activity may offer some insights, a one-to-one mapping between brain activity and subjective experience is not yet achievable. Even with advanced neuroimaging techniques such as fMRI and EEG, diagnosing depression remains a challenge. One potential approach to narrowing down the search space is to focus on behavioral measures, but the interpretability of the results may be limited, and you cannot just widely speculate on pet theories about the brain based on such behavioral measures.

One of the things I have suggested is that all of the resting state analysis should be happening on data gathered while participants are completing a task. Some proponents of resting-state activity may argue that the drug's effects on intrinsic connectivity should manifest across any task manipulation. But in my view, if an effect is too weak to persist when stimuli are introduced, it is unlikely to be a result of drug-induced changes in intrinsic connectivity. The consistent observation of decreased resting-state connectivity of the default mode network across various drugs suggests that this effect may not be as interesting as previously thought. This effect may be related to the fact that the default mode network is more active during internal mental processes, such as representing the world internally when attention is not directed outward. However, under the influence of drugs, attention may be periodically redirected to the external environment, even with alcohol. This may cause brain activity similar to those observed with stimulants and psychedelics. Therefore, failing to control for these external factors may lead to inconsistent results, as seen with various ketamine trials: three studies showed decreases in default mode network connectivity (Bonhomme et al., 2016; Scheidegger et al., 2012; Zacharias et al., 2020), two studies showed no effect (Mueller et al., 2018; Niesters et al., 2012), and the study with the largest sample size of fifty-three participants found increased default mode network connectivity. For me, these mixed results indicate a null effect.

Has the enthusiasm for the Default Mode Network (DMN) in psychedelic research mirrored the broader hype in Cognitive Neuroscience and Psychiatry for its potential diagnostic value? Many researchers viewed the Default Mode Network as one of the most reliable markers of depression (e.g. Scalabrini et al., 2020). Would you balance your critique without entirely dismissing the DMN's relevance as a therapeutic marker or the study of cognition?

> “By focusing solely on resting-state activity, researchers can manipulate and interpret brain data using unlimited mathematical techniques to support any conclusion they wish to make.”

From my perspective, the exclusive investigation of the brain in a resting state is not truly representative of Cognitive Neuroscience because it fails to consider cognition. The association between depression and Default Mode Network (DMN) connectivity is also questionable due to inconsistent findings. While the meta-analysis by Kaiser et al. (2015) found that patients with depression exhibit increased DMN connectivity, a study by Yan et al. (2019) observed reduced connectivity across a 1300-participant sample. Although methodological differences could account for the discrepancy, the main issue seems more fundamental to me. By focusing solely on resting-state activity, researchers can manipulate and interpret brain data using unlimited mathematical techniques to support any conclusion they wish to make. Russel Poldrack (2011) previously pointed out the problem of reverse inference in non-resting state data, where brain activity during a resting state is used to make assumptions about the mind rather than the other way around. Moreover, I believe this approach discredits the field of experimental psychology by ignoring behavior. But there is plenty of blame to share, as it is reasonable to suggest that some cognitive psychologists aim to enter the field of neuroscience in this manner, all while scrutinizing clinical psychology for lacking enough rigor.

How do you respond to the argument that resting-state assessments provide a more naturalistic evaluation of brain activity under the influence of drugs, especially considering these substances can lead to significant disengagement from cognitive tasks and cause substantial fluctuations in attention, making standard behavioral methods challenging to apply?

There are a few different aspects to consider here, but I think there is a general reluctance among many researchers to use behavioral tasks, due to a prevailing attitude that participants, under the influence, lack engagement and interest in the task. I generally disagree with this view. However, it is essential to acknowledge the limitations of relying exclusively on standardized methods. Cognitive batteries in psychedelics studies, such as those by Pokorny et al. (2020), often indicate cognitive impairments, but these results may not be very informative on their own. Instead, a more in-depth examination of specific tasks related to attention, memory, and other cognitive processes is necessary, as well as studies that directly compare different drugs, like dextromethorphan and psilocybin. For example, Barrett et al. (2018), reveal intriguing distinctions: dextromethorphan significantly impairs episodic memory encoding with little effect on working memory, whereas even the lowest dose of psilocybin impairs working memory but minimally affects episodic memory. These findings challenge the idea of generalized disengagement and highlight the need for a nuanced understanding of the cognitive effects of psychedelics. If task engagement drops significantly, it becomes evident through participants' responses, and monitoring reaction times and response accuracy can ensure continued engagement. We must delve deeper.

For instance, attention is not a single concept; psychedelics may enhance certain aspects of attention. Steve Luck demonstrated that individuals with lower working memory capacity sometimes remember non-cued stimuli better than those with higher capacities (Luck et al., 2002), due to fluctuating attention that increases the likelihood of including the non-cued stimulus in their mental workspace. Although this may impair overall working memory, it can lead to better performance in some cases. Investigating this effect with psychedelics, especially serotonergic ones as opposed to substances like dextromethorphan, could be intriguing. Attention encompasses various types and levels, such as covert versus overt, or bottom-up versus top-down. A study by Spitzer et al. (2001) showed that the pop-out effect, a low-level visual effect considered a type of attention, was enhanced by the R-enantiomer of methylenedioxyethylamphetamine (MDEA) but not the S-enantiomer, suggesting a specific type of attention enhancement. Although this finding requires replication, it opens new research avenues. When examining episodic memory impairment, the issue is complex. A deeper look into the data suggests that cortical-dependent memory encoding might be enhanced under certain conditions. In summary, to understand the peculiar phenomena where both high novelty and familiarity occur under the effects of psychedelics, we need to look beyond the surface and move away from overarching theories that explain little.

How do you view the use of experience sampling methods during resting-state (Fell, 2013; Gonzalez-Castillo et al., 2021; Heimann et al.) to explore the mapping of these experiences more intricately via a more fine-grained qualitative methodology?

Utilizing experience sampling techniques may probably produce better data, but you would probably need to probe people about 100 times an hour to get reliable samples to see what is going on. It also introduces the problem that participants are going to be expecting the next probe, so I think showing participants a standardized stimulus, such as a movie, would already be better. I still think there is something to be said about getting just good behavioral results to make an educated reverse inference of how people can still perform certain cognitive functions under the influence of a drug, but perhaps by using the brain in a different way than usual.

Your critique of psychedelic research appears to focus on skepticism towards subjective effect measurements and a preference for more objective behavioral paradigms, reflecting your background in cognitive psychology. You argue that relying on subjective questionnaires tends to validate preconceived notions, a practice you have (in public lectures) termed as conducting "Duh"-science. But considering the tradition in psychology to start with folk psychological understandings before advancing to scientifically validated measures, is psychedelic research still navigating early stages, where testing intuitive notions about drug effects is a necessary step?

My problem with current research is that people are not even testing folk intuitions, but rather coming in with biases from their drug-using experiences. I am sick of people acting as if they have never experienced the drug's effects. However, when you look at the things that people choose to measure, it is clear whether or not a researcher has tried certain drugs.

Investigating obvious subjective effects, such as mystical experiences or whether music sounds better under the influence of drugs, makes it clear to participants how the researchers expect them to respond. Various cultural tropes will also lead them to respond in this manner. That is why I have more faith in cognitive psychology because researchers can set up an experiment in which they stand to lose sometimes, but if they can still show an effect, it makes it all the more interesting. Based on the Mystical Experience Questionnaire scores, I would bet that many recreational drugs could induce mystical experiences at high doses. Perhaps it is useful to confirm these facts, but would you expect anything different? Despite the 'psychedelic renaissance', the reliance on predictable subjective measures leaves me questioning what new insights, if any, we have gained about the psychological impacts of these substances.

“ I have definitely not gained any new insights about the mind by researching [psychedelic] drugs. … what have you learned? I am guessing not much. ”

I have definitely not gained any new insights about the mind by researching these drugs. So let me ask you, what have you learned? I am guessing not much.

I will aim to provide a diplomatic answer to that question. Although I largely agree with all your methodological criticisms. particularly because many of your concerns seem reminiscent of my favorite term, "Neurophrenology" (or blobology), which involves trying to directly map mental phenomena to specific brain regions (Anderson, 2021; Friston, 2002; Huskey, 2016; Kosik, 2003). At the same time, I would still like to counter the polemical nature of the statement.

Firstly, because scientific discoveries rarely occur in “Eureka” moments, but rather through incremental puzzle-solving, and I would not claim that psychedelic research is exempt from this or holds any kind of exceptional, paradigm-shifting status. However, from a historical perspective, I do think that psychedelics have significantly contributed to the modern understanding of the brain, at least regarding its neurochemical basis. David Nichols (2013) articulates this better than I can, explaining that research on LSD played a significant role in elucidating the role of serotonin in the brain, leading to a deeper understanding of the neurochemical basis of the mind. The discovery of serotonin's function in the mind laid the foundation for a whole new field of psychopharmacology aimed at elucidating the neurochemical basis of the mind. Moreover, this research helped alleviate social stigma towards parents of children with autism or schizophrenia, countering the blame placed on refrigerator mothers' who were accused of withholding emotions from their children. So, in that sense, psychedelic research has already revealed a lot about the mind, albeit perhaps not through the current renaissance.

Moreover, I still think that Huxley's description of psychedelics as 'mind-revealing' has shaped current research in a meaningful way and that both the hypotheses around thalamo-cortical gating (Vollenweider & Geyer, 2001) and the relaxation of top-down constraints (Carhart-Harris & Friston, 2019) present a genuine attempt to reinterpret the notion of psychedelics as mind-revealing in a way that is conducive to scientific investigation and testing. Even though I acknowledge the caveats of the REBUS model, I argue that the scientific disagreements it provokes are valuable because they allow us to conduct experiments and build theoretical models that become more precise over time. Similar to how psychology begins by verifying folk intuitions, I think we can start with a broader theory and refine the details as we advance our understanding through encountering discrepancies.

Furthermore, I would like to advocate for the exploration of bizarre and profound subjective phenomena, as illustrated by the historical example of Hans Berger. His investigation into electroencephalography (EEG) was motivated by a near-death experience that led him to believe he had experienced telepathy (La Vaque, 1999; Millett, 2001). Despite his initial aim to explain the neurophysiological basis of telepathy, Berger instead discovered alpha waves. I believe this principle applies to curiosity about strange phenomena, such as hallucinations of entities, in that they may reveal intriguing research avenues, provided the research is conducted rigorously.

Despite my agreement with many of your criticisms regarding the use of subjective measures in current research, I believe entirely dismissing subjective data would be a mistake. Valuable insights can be gleaned from comparing the phenomenology of different states of consciousness, thereby enriching our understanding of cognition. For example, examining the qualitative aspects of various hallucinations can serve as a valid starting point for identifying differences in their cognitive and neurochemical underpinnings. Furthermore, your critique of subjective effects resonates with a long-standing debate within cognitive psychology about the reliability of introspective reports. Jack and Roepstorff (2002, 2003) emphasize that introspective evidence, crucial for understanding consciousness and subjective states, informs all aspects of cognitive science, including research and hypothesis generation. They advocate for its integration with behavioral and physiological data to gain comprehensive insights into cognitive processes. In the same vein, I challenge the notion of conducting cognitive science or psychedelic research entirely without introspection, partly inspired by discussions on 'cognitive phenomenology'—the idea that certain cognitive processes possess a phenomenal quality. Addressing phenomena such as the intense feelings of insight reported by people under the influence of psychedelics can serve as a basis for investigating corresponding cognitive mechanisms. Therefore, I advocate for an approach where cognitive and subjective measures mutually inform one another.

This brings me to my final question: Do you still remain critical of the idea that understanding experience from a first-person perspective, when approached with reflective intention, could enrich scientific practice?

In my view, your first point about serotonin is more about the brain than the mind. While I think there's something to that, take, for example, the discovery of 5-HT2A receptors on glial cells. That by itself wouldn't tell us much (yet) about the mind.

Regarding Huxley, I am not sure what the term "mind-revealing" has to do with thalamic gating or REBUS (though his "reducing valve" analogy certainly alludes to the former). I think research based initial intuition is fine, but we can not forget the plethora of foundational knowledge in psychology and neuroscience, which is why I tend to like the thalamic gating model, as it respects the known circuitry of the cortex, basal ganglia, and thalamus (though it could use more behavioral tests and refinement in terms of incorporating efferent copies).

> "To be clear, I have never claimed to be against introspection, and I am pretty unimpressed by traditional behaviorism."

Motivation by personal interest or experience can be a powerful tool for answering research questions, but we always have to remain aware that such motivation can also drive us to negate answers we do not like. To be clear, I have never claimed to be against introspection, and I am pretty unimpressed by traditional behaviorism. In fact, most cognitive measures typically rely on some form of introspection. What I find uninformative are studies that simply aim to find some obvious subjective measure (e.g., THC makes people hungry), as they are clearly prone to confirmation bias. I think psychedelic research in general needs stronger inference, and to make subjective measures more informative. The framing of questionnaires needs to be changed up in an effort to either refute or strengthen an effect across multiple experiments in a single publication, similar to how studies in cognitive psychology might contain multiple experiments to confirm an effect.

But salami slice or perish I suppose.

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# An interdisciplinary journey into consciousness research

## An interview with Juan González

By Matthieu Koroma

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### Abstract

In this interview, Pr. González and Dr. Koroma discuss the purpose and the challenges of an interdisciplinary study of consciousness. They do so by first tracing back the history of ALIUS, a 15 years-old international association dedicated to the scientific study of the diversity of consciousness. By discussing how we can incorporate views from multiple disciplines, contexts, and methodologies in our understanding of consciousness, they offer insights into the role of science and philosophy. They further address the mystery of consciousness, the place of first-hand subjective experience and empirical data in modelling consciousness, and how we can navigate among multiple definitions of consciousness and related notions associated with it like sentience or the spiritual realm.

***keywords***: *consciousness, interdiscipline, multidimensional models, psychedelics, neurophenomenology, experiential knowledge, sentience*

**Thanks a lot Juan for accepting this interview together. I would like first to come back to the history of ALIUS. When does it date back to and how did it begin?**

First I want to thank you, Matthieu, for this opportunity to express myself, and to ALIUS as well. I would say ALIUS dates back to 2006. At that time, I was having my sabbatical year in Paris and was invited by Jérôme Dokic at the École des Hautes Études en Sciences Sociales (EHESS) to deliver a series of seminars there for the 2005-2006 school year. The seminars delved into the subject of hallucinations, in the context of philosophy and cognitive science during which I met Alexandre Lehmann, a young and brilliant researcher who attended those seminars and who was very interested in the subject. We had a good chemistry since the beginning and kept in touch. Later on, next year, he introduced me to Guillaume Dumas, another brilliant student, who was also very interested in modified states of consciousness as well as trance induced by music and visuals, just like Alexandre. Alexandre and I organized a very successful workshop in 2007 on hallucinations and other modified states of consciousness and ever since the academic and public interest in those subjects kept on gaining momentum.

And that is how it all started. Then, with Guillaume, the three of us organized in 2008 an interdisciplinary Colloquium. Given its success, we started talking about founding an Association that would allow us to organize and find financial support in the future for the growing events. And so, the Association for the Transdisciplinary Research on Hallucinations and other Modified States of Consciousness (ARTHEMOC) was born as a non-profit organization. Over the years, we got a great response as people from University Paris 5, Institut Jean Nicod (IJN), Ecole Centrale, University Paris 6, the Relais d'Information sur les Sciences de la Cognition (RISC) and other institutions got involved. We started growing and the second Colloquium was held in 2009, with an even bigger success than the previous events.

Then, in 2011, I met Martin Fortier, a remarkable young man of unparalleled genius, who turned out to be a great injection of energy for ARTHEMOC, for he joined its staff with great enthusiasm. However, he had slightly different ideas and outlook on how ARTHEMOC should work and develop. A point in case is that he considered that we were too liberal and too open, because we also included shamans, independent researchers and artists to the events as speakers. Sometimes I think he was right, because when you have a large pool of diverse speakers, you go in different directions and it is hard to find a common ground for all of them and to keep high scientific standards. Yet, Martin wanted to be part of it and he joined the Association, though he pledged to have it more focused and restrained in the future. In 2015, we had our third Colloquium, organized by Guillaume and Martin, an event that started to already reflect Martin's pledge, for the better.

In the meantime, ARTHEMOC became ARTEMOC (without the H), because the idea was not to emphasize hallucinations in particular anymore, but rather the full palette of modified states of consciousness. Then other young and very qualified members invited by Martin and Guillaume joined ARTEMOC and, finally, in 2016, ALIUS was born. Let me explain. On the one hand, Alexandre Lehmann had left for Canada, and I was living in Mexico, so we could not be in France to take care of things locally. On the other hand, there was a push to adopt English as the common language for the Association and to open it to international members. So basically, we let new people take over and bring it up to a different level and direction. We were all happy about it because the bottom line was that modified or alternative states of consciousness were being studied and so with high quality standards. That is when you became part of it, Matthieu, if I am not mistaken. We were very open to letting serious people in with broad interests on the diversity of consciousness. I myself still work mostly on hallucinations. I am very happy to see that yourself and the rest have done a great job by bringing in very professional and capable people; I think ALIUS is thriving.

So do you think that some of the early objectives of ALIUS—for instance, to promote more research dedicated to alternative states of consciousness—have been attained ? Do you think there is still a way to go for the study of these states so that they can be more fully recognized?

It is hard to answer that. I do not live in France anymore and I do not know really how the scene has changed during the 15 years plus since all this started. There was some kind of resistance or caution, back then, to let these subjects be studied in academic settings. My feeling is that things have changed, for the better. In that sense, I feel there is more openness in institutions and researchers, and even in the public at large nowadays, and ALIUS/ARTEMOC/ARTHEMOC must take some credit for that. But I think France is still a rather conservative country, generally speaking, so we should not expect major breakthroughs in this respect.

Fortunately, another good decision was to open ALIUS internationally. You see, ARTHEMOC and ARTEMOC kind of took pride, to some extent, in being special and present in the French scene, as we tried to do everything in French with some things in English now and then. With ALIUS, and that is also part of Martin's contribution, it was decided that everything was to be done in English. This gave the Association a new push in the right direction, as more people interested in altered states of consciousness were reached out. I think the objectives in that regard have been reached, and rather fast, I would say.

On the other hand, from what I follow in the discussions with the current ALIUS board, some members are favoring more openness. It is a kind of *deja vu* for me because one of the questions at the beginning of ARTHEMOC was whether we should be open to let non-professional researchers and other non-academic people in. In that sense, I do not know if the objectives of ideal plurality have been reached. At any rate, it is very hard to reach a critical mass of committed and highly-qualified researchers in a non-profit organization that studies marginal phenomena and this translates as a very delicate balance to maintain the interest and involvement of people in what we do.

The idea behind ALIUS is indeed to open the study of modified states of consciousness to its diversity not only in terms of content but also in terms of disciplines and methodologies. How is this ambition challenging?

I think what you are talking about concerns cognitive science in general. When I was doing my PhD back in the 90s, it was precisely the same issues we were discussing, especially about interdisciplinarity: what methods, goals, disciplines, subjects, themes should we adopt in order to make a productive coherent whole. 30 years later, I find myself pretty much discussing the same things.

People are still asking how we should work together when studying cognition. Are we supposed to reach like an interdisciplinary platform, a common homogeneous ground, so that disciplines are somehow erased and behind us? Or is it rather like a new set of distinct parts, you know, like a Frankenstein monster that is to be shaped? I have no short answer for those very interesting and actually passionate questions. Both in cognitive science and consciousness research we have indeed different methods, disciplines, concepts and approaches involved, even within one same discipline we find a huge inner diversity (think of philosophy, psychology or neuroscience), and so in ALIUS we have different approaches to consciousness too.

In my opinion, the main challenge here is what I will call 'research management'. Here in Mexico, as the head of a research center on Cognitive Sciences, I see the beauty but I also endure the challenge of having anthropologists, neuroscientists, linguists, philosophers, and psychologists working together. In our own research center, we strive to work at 3 different levels simultaneously. The first one is the individual level, where everyone does what they please, following their own interests and subject matters. The second level involves the disciplinary area of research. Every researcher belongs to at least one of the five areas of the center. In said areas, we have to talk to each other and interact, which is already a challenge because, like I said, even an area like psychology can be huge, a little universe in itself, and it is the same for neuroscience and so on. In addition, we are expected to publish together and organize monthly seminars within our area. The third level, and the most challenging one, is the interdisciplinary level. To give you an idea, every week students and researchers gather in an amphitheater to see and hear the students' thesis progress. It is a very eclectic crowd, because students and researchers come from very different disciplines. Feed-back from the audience to the students is expected, and spontaneous discussions are welcome.

The goals are clear in this third level, but attaining them is the problem. It is a matter of practicing and believing. It is like dieting or exercising. We know that it is good for us, but we don't always follow through. The real challenge is the practice. We cannot take interdisciplinarity as a given, nor its approaches for granted. It is something that we must build, and we are still working toward that end and cannot just relax and think that everything is done already. Developing interdisciplinarity implies being willing to sacrifice a bit of your own disciplinary identity in order to reach a middle ground, and this is challenging. It has to do with personal adaptability and cognitive plasticity, and with the interest or motivation you may or may not have to reach for others out there. In this third level, the first step forward requires attitude and maintaining the channel open, so to speak, and the conviction that we enrich and enlighten each other from different perspectives on subjects that we all care about. The second step is to develop a conceptual platform and, ideally, a methodological platform too that allows us to understand and cross-pollinate each other. In my experience, I have found that at very small scales. I do not know to what extent we can develop it further in a research institution or at bigger scales. Could it be that, for interdiscipline, we need discipline, on both accounts of the term?

> “The real challenge is the practice.
> We cannot take interdisciplinarity as a given,
> nor its approaches for granted.”

I think that it is one of the ambitions of Martin and other people of ALIUS to build a multi-dimensional model of the conscious space allowing us to put on the same map different states and models of consciousness (https://www.aliusresearch.org/about.html). The rationale is that our understanding of one state of consciousness can be enriched by comparing it to others and combining different levels of studies and methodologies (e.g., Zamberlan et al., 2018). More recently, Chris Timmerman and colleagues have for example developed a neurophenomonological framework on non-ordinary states of consciousness encompassing neuronal and phenomenological approaches on meditation, hypnosis and psychedelics (Timmermann et al., 2022).

Building multi-dimensional models is still in development and there remains a lot of questions about the dimensions and the kind of multidimensional models that can be built to map conscious states (Bayne et al., 2016). Despite good progress over the past decade in enriching these models (*e.g.*, Sergent et al., 2017), there is still no consensus about the right level on which we should look at these phenomena, how to incorporate phenomenology with neuronal data, and how anthropology can be included in those models. Do you see any limits to such ambition?

Of course, there are all kinds of limits. There are, for instance, top-down institutional limits. We have all experienced a paper being rejected by editors saying that it is not for their journals, or we have seen university departments with people who feel uneasy about their institutional or disciplinary identity: we need to find a niche for interdisciplinary studies. And there are conceptual, methodological, sociological and psychological limits too. We need to build models and even meta-models on consciousness, but it is a very hard thing to do if they are to please everyone.

Philosophers and philosophy itself have been a federating force in the development of traditional cognitive science and in putting together models of consciousness. However, some researchers are unhappy with this, as they think that philosophers miss out on some important experimental aspects of research, like statistics or the molecular level. Perhaps they are right in saying so, but philosophers would also say that you need to consider other levels or aspects when it comes to studying consciousness, like emerging levels or experiential aspects that you leave out when you try to be reductionistic. So different disciplines or approaches do not necessarily tackle the same epistemological problem. I think we need to be multi-level, multi-dimensional and interdisciplinary, but the problem is the road map to get there. As for me, I gave up trying to think of cognitive science in the singular and I now believe more in a family approach of different aspects. I can say the same about the study of consciousness.

From a Wittgensteinien perspective, I would say that we need to give up our longing for absolute and/or definitive definitions and rather bring in the idea of language-games. Martin Fortier, just as most of us, was not pleased with Arnold Ludwig's famous definition of an "altered" state of consciousness (Ludwig, 1966). Nowadays, I do not think that we have a better definition. We just have alternative definitions. Martin himself was struggling to provide a definition—that I am not happy with just as I am not happy with my own definition either. I think cognitive science and approaches to consciousness resort to different language-games for studying different aspects of cognition and consciousness, respectively.

Jumping back to this idea that we enter language-games, I wanted to discuss about the notion of "state" of consciousness. I have been questioning myself what qualifies as a "state" of consciousness, especially in

psychedelic states, where you have such a plasticity and transformation of your experience. We could frame it in terms of a transformative experience where you learn and transform yourself, and maybe we could gain a better understanding of these states with a more dynamic approach.
What makes a "state" ?

Traditionally, a state is something that has a beginning and an end in time. That definition already poses a big problem when it comes to consciousness, because I do not think that you have that when applying it to 'states of consciousness', and even less so when applying it to 'altered states of consciousness'. William James spoke about the stream of consciousness. I like this image because I think of a river in which you may look at different currents or threads and depths. Plus, there is no one single thread, going from the beginning until the very end of the rope, just like there is no identifiable set of individual currents that together supposedly make up the river as a whole.

Of course, to make the work with those notions simple, we need basic definitions that are certainly not miraculous but rather just working definitions that help clarify what is at stake. So I think a state can be defined just like that, as a (mental or physical) steady configuration that has a beginning and an end, and then it changes. I like what you say about dynamics and there are modeling approaches to dynamic states or to the dynamic operation of brain activity that can be useful here. Perhaps the state or activity should be conceived more like a gradient, which in turn is recognized to be part of the whole stream or part of a larger set of multilevel states; within such a gradient, we could distinguish benchmarks, because there are peaks and valleys, but also phase transitions. I think these are good ways to visualize psychic organization and the concomitant information, and also useful to understand the psychedelic experience or whatever your experience is about.

How can psychedelics then inform on the neurobiological and sociological constraints that make a conscious state?

A very important part that philosophers, including myself, tend to forget when it comes to cognition is biology. We need to know the most we can about our cognitive system, including the brain and the nervous system, to build our own theories. I have always thought that phenomenology is constrained and broadly determined by neurobiology. We just cannot think of experience without being embodied and embodiment obliges you to consider the neurobiology of the system, together with its functions, evolution and ecological relations. Then you speak of sociological constraints. Cultural, linguistic, conceptual constraints all come together, like Russian dolls. The fine tuning resides in putting everything in its own place and in the right proportion when it comes to study consciousness and psychedelics: biology, psychology, sociology and philosophy.

> “I have always thought that phenomenology is constrained and broadly determined by neurobiology. We just cannot think of experience without being embodied and embodiment obliges you to consider the neurobiology of the system, together with its functions, evolution and ecological relations.”

ALIUS has always had the position to stay away from the legal aspects of modified conscious states, for example concerning the use of psychoactive substances or psychiatric conditions. One of ALIUS tenets is that a better depiction of these states of consciousness in society can be achieved by reinforcing the scientific knowledge about the diversity of consciousness. What are the political consequences to you of raising a more informed awareness on the diversity of consciousness? If we talk also about the diversity of cultures in which these modified states of consciousness have been cultivated, could we consider some risks, for example, of cultural appropriation? How can we benefit from cross-cultural talks in the study of consciousness?

There has always been a need, for human beings, to deliberately alter our own consciousness, like most of us have had the experience of being drunk, or riding a roller coaster. It is pretty much an anthropological fact that, at some point, we need to explore and experiment with our own consciousness. Even other animals do change drastically their behavior by intoxicating themselves through plants or other means. I think that kind of need is universal. There are traditions like rituals of initiation that are culturally inscribed and designed to respond to that need. There is a true cognitive need for pushing limits and answering existential questions that we have always had as humans.

In traditional societies, native people in different parts of the world have recognized those needs and made them part of their lives and healthy development. But I think Western society has very often looked the other way, because of ignorance, cultural bias or fear of deepening the knowledge of our own consciousness. Then add the political aspect, which has lagged behind scientific knowledge on the subject of psychedelics for a long time. However, in the last 5 or 10 years, we have seen some comebacks and breakthroughs regarding psychedelics, mainly because of their clinical application. It is no surprise to see, in our dysfunctional materialistic societies, how mental illnesses such as depression or anxiety have risen in the last decades, with close to pandemic proportions, especially in the young population (https://ourworldindata.org/mental-health). During the Covid pandemic, this has gotten worse (https://www.who.int/news/item/02-03-2022-covid-19-pandemic-triggers-25-increase-in-prevalence-of-anxiety-and-depression-worldwide). There is clearly an urgent need for taking care of our mental health, but our Occidental approaches by default have been rather inadequate, I am afraid. On the one hand, standard medical approaches tend to focus on symptoms rather than on root causes, on the other hand, there are biases and even taboos regarding 'alternative' medicine, such as shamanic healing.

I do not want to idealize or romanticize native people, but I have seen and learnt ways they have to raise children and to overall live in a healthy natural and social environment. For one thing, they form communities and are attached to collective values, for another, they are connected to nature and to the natural cycles of life, all of which have a positive impact on their own physical and mental health and development. In contrast, in Occidental urban environments, it is very hard for young people nowadays to find their place and balance in a individualistic world, where success, money and competition are paramount values, where healthy habits are hard to implement and maintain, where social media, video games, virtual reality and the meta-verse occupy a good part of their mental space, and so on. In these environments, being healthy in a rich or holistic sense (including the spiritual, psychological, emotional, physical and social dimensions) is very challenging, and I don't think we are doing a good job, overall speaking, in providing our youngsters with the means to develop a healthy and meaningful life.

Rick Doblin, then President of MAPS, told us in the first colloquium of ARTHEMOC in 2008, that clinical aspects of psychedelics are important because they help and validate the scientific research on alternative states of consciousness. I think that is very welcome, but I do not think we have to stop there because psychedelics and research on consciousness go far beyond the mere instrumental and clinical. They are very important to heal psychic disorders, but also, if anyone cares, to know ourselves and find meaning (again) in what we do, in what we are, an aspect that we often lack in modern life nowadays. Together with this, we need a larger epistemological framework that includes other people's cosmological visions, like the native people here in Mexico. Fortunately, we still have a large group of native people that are willing to show us how they live and to teach us their wisdom. This calls for social reconfigurations. I do not know if you would call it a tribal organization, but definitely it is about getting together, trying to inject more sense in what we do just by relating to one another and finding new ways to relate to the environment in meaningful and ethical terms.

Do you think then that more connections between consciousness science and other approaches to conscious experience like artistic or ritualistic can help? What is the place of science in making sense and meaning of our own experience?

Well, it depends on the scientific person we are thinking about. If you are broad minded and you reckon that science does not and cannot encompass our full understanding of reality, then I think that science and scientists can do a great job in building, validating or reinforcing connections to other worldviews. But if you are very limited in your outlook on reality, and either implicitly or explicitly dismiss other people's way of looking at the world, such as the time-tested wisdom of native people or the view and practice of confirmed artists, then I think you miss out and cannot connect. It must be admitted that artists often bring new concepts and ways to look at the world and these are definitely worthwhile.

The only problem with liberal postures like mine is that, when you let many things in, then you run the risk of making a mess of epistemology; with a larger and pluralistic epistemological framework, you need special caution. We need to put things in their appropriate place, we need to be systematic and methodical. I think it is good that scientists help us to organize our understanding of reality with rigor and discipline. But science itself is not enough. So again, it is about an epistemological balance, neither being too open to let anything in nor too rigid and narrow to leave important things out. I think that that framework is within our reach and the first step consists in educating scientists, making them aware of their own epistemological limits. The second step consists, in my opinion, in bringing experience to the fore, both as a concept and a feature of reality. This is *terra incognita* for most scientists. In any case, there is a lot at stake, and scientists have a big responsibility, for people in general look up to them and heed their advice.

> “ If you are broad minded and you reckon that science does not and cannot encompass our full understanding of reality, then I think that science and scientists can do a great job in building, validating or reinforcing connections to other worldviews. ”

Do you think that there are some methodological aspects here that are specific to consciousness and phenomenology to discuss, notions like experiential knowledge that is the kind of knowledge gained from our own experience of the reality as we live it? Do you think considering this kind of knowledge is a way for scientists to be more modest in their objective outlook on reality and emphasize more subjective approaches of how people live and make sense of the world?

Like I said, I think there is a great potential gain in the collaboration between science, epistemology and phenomenology. As a personal anecdote, the first time I drank Ayahuasca in Peru, I had a revelation. It was an epiphany, because, among other things, it allowed me for the first time to feel at ease between science and the spiritual realm, between my own experience and objective descriptions of reality. I work in a university and for me that was very relieving: to finally put together, in a coherent, non-exclusive picture, those two important aspects of my life. To be sure, this outlook requires discipline, clarity, and reflection; in addition, I think it is important not to mix language-games. In a university, we have to be scientific and be conceptually clear, organized, and methodical. But we also need to be open to new ways of looking at the world and reality. The problem again is how much is too much or how little is too little: how receptive or how dogmatic you are.

Science has a lot to say about consciousness, but I think that every scientist must also recognize that experience is part of what they are and what they know. When you undergo a psychedelic experience, sometimes your understanding expands, sometimes it is shattered or sometimes it just changes. You could even have a revelation. Philosophers of mind, and I consider myself one of them, should get to know their mind, nuts and bolts. Deliberately modifying it is a great way to know your mind firsthand. And not only philosophers of mind, I think also scientists of mind, should get to experience it in a modified way at least once in their life. Through modification of consciousness things appear in a different light. Of course, it is not an illusion nor a hallucination—that is another typical way to dismiss it. First-hand experience, like Francisco Varela would say, is cognitively paramount and not replaceable by anything.

We have not talked about Shamanism, and I know that in Europe Shamanism is basically lost, or very rare in the traditional sense. However, we could resort to the Shamanic outlook on life to reconnect our own consciousness with a deeper meaning of what being alive means, what being on this Earth, on this universe means, not only in a conceptual way, but in an experiential way too. The credo of the phenomenologists? First-hand experience. Experience pervades everything and is at the center of everything we think or feel. We need to put experience in its right place.

> “Philosophers of mind, and I consider myself one of them, should get to know their mind, nuts and bolts. Deliberately modifying it is a great way to know your mind firsthand.”

### What about the notion of sentience?

Sentience is a concept that has not been talked much about and that came up to me a couple of years ago in a course I taught and a thesis I supervised. It is an important concept because it helps us in deciding whether bullfighting, circuses or abortion is right or not. We make very important decisions based on that concept. So sentience is a good subject of research by which consciousness would at first seem more tractable. But it is a very difficult concept to deal with when it comes to the origins and nature of sentience. It has to do with the hard problem of consciousness, because I would fancy that sentience is a form of primitive consciousness, a sort of angular stone for consciousness, from an evolutionary perspective. It is also something that carries the minimal sense of consciousness, like when you are in coma and coming back from it. It is about feeling, feeling yourself, feeling alive or feeling cold.

> “Sentience is a form of primitive consciousness, a sort of angular stone for consciousness, from an evolutionary perspective.”

The use of the term in research right now is a bit diverse. There are different conceptions, on the one hand related to the subjective feeling of conscious experience and related to ethical concerns (Almada & Linnell, 2021), and on the other hand, in the context of active inference, related to the ability of a system to interact with its sensory environment, *i.e.*, “the sense of being “responsive to sensory impressions”” (Friston et al., 2020). The implications of using one or the other definition fueled some confusion and debates about the extent to which these conceptions are sound in terms of definition and methodologies (*e.g.*, Kagan et al., 2022).

Until now, I have thus not seen a consensus emerging on the use of the term sentience. It remains open to what extent painful experience in vegetative states or minimally conscious states comes with an ability to be responsive to sensory stimuli. In anesthesia, it remains unknown whether responsiveness to bodily interventions are encoded and stored as a trace in the body or the brain in a way that could qualify as a conscious or unconscious experience. So it is a very interesting concept to discuss how certain aspects of experience and consciousness might be tractable in terms of mechanisms and definitions.

Evan Thompson has a recent paper asking: "could all life be sentient?" (Thompson, 2022). I found that it was a very good article, except that in order to make it tractable, he defines sentience by hedonic value, like pain or pleasure, or affective valence. I think that sentience goes even deeper than that, and that there is like a minimal, more compact concept underneath. Before we consider responsiveness to pain or pleasure, to affective valence (feeling x), or to sensory stimuli, like you mention, we should consider the capacity or possibility to feel *tout court*. Here is where metaphysics joins evolutionary theory, cognitive science and consciousness studies. Sort of Kant meets Darwin, Varela and Merleau-Ponty on cognition and consciousness.

There is another case of conceptual confusion that I am curious about. When we talk about theories of consciousness like panpsychism, we can also ask to what extent all life or even matter is conscious (Tononi & Koch, 2015). I am wondering here if there was a confusion between the spiritual, or the immaterial aspects of things, and consciousness *per se*. We can make the case that they are interrelated at some level depending on theories of consciousness. But I am wondering how you would differentiate both, and whether you see a similar case of conceptual confusion arising here?

I had not thought of it, you know. But I can see what you mean, and of course, it is a huge confusion. On the one hand, when you talk about sentience, you need to come back down to the evolutionary origins of life, what were the material conditions for that beginning. Biology can tell us a lot about that. The spiritual or animist part has nothing to do with it. You have to cook it in a different pot. When you are in a shamanistic environment, a ceremony, or in a ritual, you may feel the spirit of Peyote, or Ayahuasca, or the water, for instance. You are able to sort of communicate with it—in the sense of a spiritual communion with that spirit—and feel yourself possessed by, or connected to, its "personality" or "wisdom" or "greatness" (it is hard to put into words this type of experience. What cannot be said, we should indeed pass over in silence, as Wittgenstein wisely recommends us). But it does not make sense to say that these plants or the water are conscious or wise from a scientific perspective. I think it is possible, and desirable, to keep those two language-games and contexts separate.

There are several other points that you wanted to discuss, ranging from perception to interdisciplinarity, evolution and ecology.

I think discussing perception would take us too far down the road because I am passionate about it and it is a field of expertise for me, especially visual perception. The fascinating point about perception is that it ties experiential issues with conceptual ones, as well as the neurobiological and functional aspects with the cultural aspects of cognition. Perception is a great subject matter for working on cognition and the diversity of consciousness. It also relates to ecological and evolutionary psychology. We need to take ecological and evolutionary aspects seriously to understand our own make-up and our own cognition across time and space in a more relational way. Sometimes, I find accounts of cognition to be too reductionist because they leave one aspect out or the other. Of course, it also ties in with the interdisciplinary method. It is something that we have to keep working on and I celebrate that ALIUS has this spirit because bringing these different aspects together is very challenging.

Do you think interdisciplinarity is especially important for a research topic like consciousness? Would a theory of consciousness, if there is one possible, be expected to emerge from interdisciplinary dynamics?
Or should we look maybe for a driving discipline to bring something that would give an entirely new way to look about consciousness?

The problem I see with consciousness is that oftentimes we tend to reify it.

We often no longer know whether the problem is real or just a mental construct. I try to keep a deflationist approach to consciousness in order to keep things manageable. I think scientists and philosophers need to keep things compact and as clear as possible, and be very sober and parsimonious. In that regard, I do not foresee a breakthrough or something like that in the near or far future, because I think that, to a great extent, it is a conceptual rather than an empirical matter.

But there are certainly conceptual and experiential approaches to consciousness that can enrich each other. For instance, empirical study helps to validate or experiential theories of consciousness, whereas conceptual analysis may help clarifying issues and redirecting the empirical efforts of the study. You mentioned the study from Tagliazucchi comparing reports of conscious experience from psychedelics and dreaming experiences (Sanz et al., 2018). They found that more similarity exists between dreaming and psychedelic experience compared to wakeful experience. That makes a lot of sense and lots of empirical research can be done in those directions. But then again, does it inform you on what is a conscious experience to begin with? I do not see a breakthrough in the scientific approach to consciousness telling us something like Eureka, we found the mystery of consciousness.

> "I do not foresee a breakthrough or something like that in the near or far future, because I think that, to a great extent, consciousness is a conceptual rather than an empirical matter."

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# What is it like to be in weightlessness ?

## An episode with Steven Jillings

Recorded by Daniel Friedman and edited by Lena Coutrot and Matthieu Koroma

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### Abstract

In this episode, Steven Jillings reviews the recent progress on the scientific study of conscious experiences in weightlessness.

***keywords***: *Weightlessness, Free Fall, Spaceflight, Parabolic Flight, Vestibular System, Inversion Illusion*

The video and audio version of this episode are available on https://aliusresearch.org/bulletin06-jillings-podcast.html

For as long as humankind can remember, we've been astounded by the mysteries of outer space. Major advancements in our understanding of the physical world in the last few centuries have led to technological developments that enable us to travel farther and farther away from Earth's surface. Although many engineering, infrastructural, and physical aspects of space travel have been successfully established, how the human body reacts and copes with such extreme conditions was and still is much less known. Of the various aspects of spaceflight, weightlessness is probably the most remarkable difference experienced by space travelers. Because biological organisms have evolved throughout a long period of time under Earth's gravitational conditions (known as 1G), their anatomical and physiological systems have developed to work optimally within this condition. Humans, therefore, experience a profound change when in weightlessness, both in terms of unconscious physiological mechanisms as well as in terms of subjective feelings.

> Of the various aspects of spaceflight, weightlessness is probably the most remarkable difference experienced by space travelers.

Starting from the first human space mission in 1961, a new scientific research domain has emerged to unravel the ways in which the human body reacts and adjusts to weightlessness in order to ensure successful and healthy missions in space. Several approaches to studying the effects of weightlessness on the human body exist. The most obvious way to study the effects of weightlessness is to take measurements of astronauts while they are free-floating during their space mission. Such measurements, however, are restricted to either subjective reports or to measurements with relatively simple equipment that can be transported to space. A second approach is to perform experiments on-ground before and after spaceflight, which has the advantage of access to the full range of laboratory equipment, but has the disadvantage that measurements are not taken in a weightless condition.

A common limitation for both spaceflight experimental approaches is that few people go to space, leading to a slow pace of knowledge gain on the effects of weightlessness on the human body and mind. To overcome this limitation, experimental settings on Earth have been developed to simulate various aspects of spaceflight. One example is parabolic flight, which is the only way to create weightlessness on Earth. This is realized by an airplane making parabolic trajectories, where passengers experience weightlessness for 20-25 seconds during the top part of the parabola (Karmali & Shelhamer, 2008). In this setting, researchers can conduct in-flight experiments, as well as before/after study designs, although the passengers are only weightless for a short consecutive period.

At the start of the free floating phase, disorientation and nausea may occur, which in some cases may lead to vomiting (Young et al., 1984). This is true for both spaceflight and the short-term spells of weightlessness induced by parabolic flight (Glasauer & Mittelstaedt, 1997). Weightlessness can further lead to visual and bodily illusions, such as the inversion illusion, where one's sense of uprightness is lost (Gabriel et al., 1967). The reason for these experiences is that the brain is unable to make sense of conflicting signals coming from the eyes and the vestibular or balance system. On Earth, our vestibular system detects head position changes, which relies in part on the existence of gravity as a vertical reference. As a consequence, humans in weightlessness are unable to distinguish up from down based purely on their senses (Senot et al., 2012).

> “At the start of the free floating phase, disorientation and nausea may occur (...). Weightlessness can further lead to visual and bodily illusions, such as the inversion illusion, where one’s sense of uprightness is lost.”




It also seems that these altered perceptions are confined to the spatial environment and not to the sense of embodiment. A study in Russian cosmonauts found that 98% of the tested cohort experienced disorientation, while no one reported on altered perception of body ownership or embodiment (Kornilova, 1997). Additionally, the physical laws governed by gravity on Earth are hardwired within our brain to facilitate predictions of object movements in the environment and to generate quick motor responses to interact with the moving object (Iodavina et al., 2005; Senot et al., 2012). Sensorimotor functions in a condition where the known laws of physics do not seem to apply will therefore be required to update and adopt new strategies, such as the trajectory of goal-directed arm movements (Gaveau et al., 2011).

Finally, studies using parabolic flight have shown that weightlessness might trigger altered arousal responses based on observing increased stress hormone release and elevated high-frequency neurological activity. On the other hand, after being exposed to several parabolas, a suppression of this neurological activity was found, suggesting that the participant has got somewhat familiar to the state of weightlessness (Schneider et al., 2018). Prolonged stays in outer space further trigger changes in astronauts’ conscious experiences. Astronauts typically have poor sleep quality due to a floating “position” during sleep (Wu et al., 2018). Accumulation of sleep deficit may then affect high-demanding cognitive tasks and cause mood changes. Despite all these operational difficulties and triggers of the brain's autonomic responses, weightlessness is generally considered to be a profound and fun experience. Sally Ride said she believes experiencing weightlessness, along with viewing the Earth from a great distance, are for any astronaut the most fun aspects of spaceflight (https://kidadl.com/articles/inspirational-astronaut-quotes).

> “Experiencing weightlessness, along with viewing the Earth from a great distance, are for any astronaut the most fun aspects of spaceflight.”

With the prospect of future long-duration missions to the Moon and Mars, it will be important to understand how different G-levels (1/6G on the Moon and 1/3G on Mars) during such missions influence conscious states and adaptability. A remarkable observation in light of transitions into different gravitational environments are the consistent reports that astronauts who have been to space before adapt more quickly to the space environment (Reschke et al., 1998). Upon return to Earth, these experienced astronauts also acclimatize quicker to Earth's gravity than their novice counterparts. Although there will always be a time window required for adaptations, a period where astronauts are most vulnerable in terms of performance, this observation suggests that human beings are capable of learning to live and work in a space environment. Nevertheless, the unfamiliarity of microgravity and other gravitational states will always have a profound effect on a vast range of brain functions, as the hardwired brain computations that have accumulated and have been optimized throughout one's life on Earth are seriously challenged during spaceflight.

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# What is it like to be asleep ?

## An episode with Matthieu Koroma

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### Abstract

In this episode, Matthieu Koroma reviews the recent progress on the scientific study of conscious experiences during sleep.

***keywords***: *Sleep, Dreaming, Lucid Dreaming, Hypnagogic Dreaming, Memory, Self, Time*

The video and audio version of this episode are available on https://aliusresearch.org/bulletin06-koroma-podcast.html

Conscious experiences during sleep have long been considered mysterious and typically difficult to assess as they can only be shared after awakening. Even the memories that we are left with in the morning are often faint and sometimes bizarre. So a question naturally arises : do memories after awakening truly reflect the type of conscious experiences that we have during sleep ? Even more dramatically, could memories be completely made up to such an extent that we actually do not dream at all during sleep? (Malcolm, 1953; Dennett, 1976). This hypothesis has been debated over the past 50 years but the scientific consensus is now suggesting that dream reports are actually faithful (Windt, 2013). One of the strongest arguments in favor of this hypothesis comes from people who have a pathological condition called REM behavior disorder. Because people with REM behavior disorder are unable to inhibit their movements while dreaming, researchers could observe their actions and found that they fit remarkably well with the dream experiences that were described after awakening (Oudiette et al., 2009).

> Could memories be made up to the such an extent that we actually do not dream at all during sleep?

So if we can trust dream reports, what do they tell us? Researchers have studied this question systematically by waking up participants at different moments of the night and asking them to describe the conscious experience that they recall from their dreams. What they found is that dream contents varied greatly depending on the time of the night (Casagrande et al., 1996; Siclari et al., 2013). While we are falling asleep, we can experience intense conscious activity that is called hypnagogic dreams. These dreams often reflect what we have been experiencing during the day. This is known as the Tetris effect, because doing some actions for several hours, such as playing the video game Tetris, causes images of this game to be replayed during sleep (Stickgold, 2000). Even while firmly asleep, our dreams tend to be composed of images and actions related to our daily activities. But as the night goes on, the more complex our dreams become.

One aspect of dreams that has drawn particular attention is our experience of the self (Revonsuo, 2005). Our subjective experience can be in the first person perspective, embracing the point of view of our body. However, we may also experience dreams in the third person, floating slightly above ourselves. We may even be absent from the dream, adopting a disembodied point of view that simply watches the dream unfold. Another focus on the dream experience has been the perception of time. While a dream report can describe actions taking place over several hours or days, we only dream for a few minutes or hours. However, a good correlation was found between the actual time spent asleep and the duration of dream experiences (Dement & Wolpert, 1958). An explanation for this fact could be that, in our dreams, we are not so surprised by transitions that involve jump in time, a phenomenon that is also found in books or movies and called narrative ellipses.

> The dreamless sleep experience has been described in particular detail by great meditators.

Finally, we can also wake up knowing that we experienced something in our dream but without being able to say precisely what it was about. While this may reflect a failure of our memory, this could also tap into the existence of conscious experiences that are lacking any content. This phenomenon is known as mind-blanking during wakefulness (Ward & Wegner, 2012) and white dreams during sleep (Fazekas et al., 2019). This dreamless sleep experience has been described in particular detail by great meditators who have trained themselves to remain conscious even after falling asleep (Windt et al., 2016).

Another trainable skill that has sparkled the interest of researchers as a tool to probe the nature of dreams is lucid dreaming. Lucid dreaming is the ability to become aware of dreaming while in a dream (LaBerge & Rheingold, 1990). A question that has been investigated is to what extent the body of the sleeper is involved when we dream. Researchers found that sleepers' respiratory rhythm slows down if lucid dreamers are asked to stop breathing in their dream (Oudiette et al., 2019). By asking different questions to lucid dreamers while recording their body and brain responses, five laboratories around the world have now succeeded in communicating in real-time with lucid dreamers (Konkoly et al., 2021). This rise in research about dream and lucid dreaming offers us novel and unique perspectives to better understand what it is like to be conscious while sleeping.

> “ Lucid dreaming is the ability to become aware to become aware of dreaming while in a dream. ”

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# On free will or the lack thereof

## An interview with Robert Sapolsky

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## Abstract

In this interview, Robert Sapolsky outlines his view on Free Will and related topics. The discussion anticipates his upcoming book *Determined: The Science of Life Without Free Will*. Various topics are covered at the intersection of neuroscience with philosophy, education and the criminal justice system.

***Keywords***: *Free Will, Human Behavior, Legal System, Philosophy, Complexity*

In your career you have worked across areas: doing empirical field and laboratory work, crafting various kinds of articles, and teaching legions of undergraduates. Also you have written several popular science books (Sapolsky *et al.*, 2004; 2005; 2017), variously bringing attention to ulcers, hormones, and learning. Recently you have appeared on various podcasts such as with Prof. Andrew Huberman (Huberman & Sapolsky, 2022), Here We Are (Mauss & Sapolsky, 2022), and Freakonomics Radio (Levitt & Sapolsky, 2021). To pick up here with this interview where some of these previous conversations have left off, on Freakonomics Radio you said: *"I don't think we have any free will whatsoever. I think we are the outcomes of the sheer random, good and bad biological luck that each of us has stumbled into"*, and on the Huberman podcast you denied that humans have even a "*shred of free will*". For clarity in this interview, can you provide your usage or definition of free will here? What are your definitions of consciousness and awareness?

These are great questions that I'm spending half my waking hours obsessing over. Philosophers/legal people define free will along the lines of, having had alternative options when you carry out some behavior, understanding that there are those options, understanding the implications of what you've chosen, having the ability to veto impulses to behave in particular ways (what usually gets called "free won't"). I come from way out in left field, in terms of how completely biological my perspective is. A behavior happens, something as simple as you decide to bend your finger. You can probably isolate the one motor neuron that initiated that action (or least a small handful of neurons). They activated because a neuron(s) upstream activated it, which did so because upstream from it...

So, let's take a neuron that has just triggered a behavior in this sense. Show me that it did so without any inputs from any upstream or surrounding neurons. Show me that it would have done the exact same thing at that moment, regardless of how much the person slept last night, their blood glucose levels, levels of various in the bloodstream around that time. Show me that it would have done the same even among subliminal cues and primes that psychologists show will alter behavior. Then, show me that it would have done the same regardless of the person's childhood, fetal life, genomic makeup. Show me that the neuron's behavior was not shaped by any of these influences, and you've demonstrated free will. This is obviously reductive to a silly extent, but you get the idea.

Philosophers and legal people spend a huge amount of time focusing on, "Did the person intend to take that action?" and in courtrooms that's one of the hallmarks of what would qualify for legal/criminal responsibility/expression of free will. And this ignores the 99% of determinative events that explain, "And where did that intent come from in the first place?" It's like judging a movie after only seeing its final three minutes.

> "Show me that the neuron's behavior was not shaped by any of these influences, and you've demonstrated free will."

Soon you will be publishing a new book, *Determined: The Science of Life Without Free Will* (Sapolsky, forthcoming), adding another discourse into the fray regarding free will (O'Connor *et al.*, 2018). To judge a book by its title in advance of reading—in the polysemous word "Determined" we can see allusion to both sides of the debate. From one view, if the outcome of a situation has been determined, then no agency can be exerted at all. In this view, a deterministic perspective, free will indeed does not exist. On the other view, for example the case of a person feeling *determined* to achieve a goal, determination seems like a canonical case where applied free will provides the motivation and grit to succeed. While surely the book itself will be addressing these questions as well—What motivated or influenced you to write this book now? What do you hope the book's impact will be?

I'm glad you perceived the (intended) ambiguity in the title. Why am I writing it now? I published this book, *Behave: The Biology of Humans at Our Best and Worst* in 2017 (Sapolsky, 2017), and did a ton of public talks about it. It basically goes through how our behavior is the outcome of events in the brain one second before, hormone levels that morning, experience in previous months, childhood, fetal life, genes, cultural evolution, all these things over which we have no control. And I'd be shocked by how many audience members, during questions, would show that the idea of there being no free will, or at least so little as to be irrelevant to interesting stuff, was novel. It seems like an obvious implicit conclusion from that book. So given those responses, I decided to make it a little more explicit.

What impact am I hoping the book will have? Obviously, I'm hoping it will solve global warming, suggest a cure for Alzheimer's Disease, save mountain gorillas from extinction, *etc.*, *etc.* Realistically, I'll be happy if at least a few people read it and find the arguments to be convincing.

What does free will feel like? Or, what would/could it feel like?

It's the most natural feeling in the world, totally reflecting this enormous human need for a sense of agency. It's something we desperately hold on to—loss of a sense of agency is at the core of the learned helplessness of depression, and of the hypervigilance of anxiety. It's so tough to take apart because we are working simply on the conscious level of attribution—it's so counterintuitive to think that an answer to the question, "Why did you do that just now?", is one that shows that 99% of what is going on in us is subterranean. Who finds it natural to think that one of the explanations for why you just did something wonderful and altruistic is related to your oxytocin levels this morning, or how your cingulate cortex was constructed when you were a third trimester fetus? Our conscious sense of agency is usually post-hoc attributions to make sense of what you just did.

> “ Our conscious sense of agency is usually post-hoc attributions to make sense of what you just did. ”

Why do people—many of them quite honestly—feel that they have free will?

The answers are embedded in the one above—because, cognitively, it is so hard to understand the influences of all these things you can't see (back to hormone levels, fetal environment, *etc*). Emotionally, because it's depressing as hell to view ourselves as nothing more than the outcome of the biology, over which we had no control, and its interactions with the environment, over which we had no control.

The cognitive stances we take can be associated with real differences in personal outcomes. While the preceding sentence was mindful not to imply that such cognitive stances play a causal role in behavior, in various literatures it is common to read about the "impact" or "effect" that beliefs have on behavior and life outcomes (*e.g.* Conversano *et al.*, 2016; Sirgy, 2021). In your view, how is our personal stance on free will associated with our behaviors and context? Where are the causal arrows pointing—what leads to what?

Something that everyone assumes is that if people stopped believing in free will, everyone would run amok, there would be no constraint on behavior, madness in the streets. Moreover, some experimental studies have shown that when you lessen someone's belief in free will, they will cheat more in an economic game, will lie more, *etc*. Three responses to that:
a) those studies haven't been replicated;
b) when you look at people who come into the study ALREADY believing there is no free will, already having struggled with the implications of that, and so on, they are just as ethical and moral as those who do;
c) this notion of running amok if people stop believing in free will is a version of, "Why not, I'm not responsible for my actions."

A very large body of literature concerns a related conclusion of, "Why not run amok, after all I'm an atheist and don't think there is an ultimate Judgment." However, when you study it correctly, reflective atheists are exactly as moral as highly religious people.

In work such as the 1971 book *Beyond Freedom and Dignity* (Skinner, 1971) and *Walden Two* (Kulman, 2010; Skinner, 1973; 1948), B.F. Skinner popularized a behaviorist perspective, which in the human setting emphasizes the role of external reinforcement in social control. Skinner wrote: "Almost all major problems involve human behavior, and they cannot be solved by physical and biological technology alone. What is needed is a technology of human behavior." What do you think such behavioral technologies could look like today and tomorrow, and how can we design systems for human benefit?

That's going to be tough as hell, for the reasons outlined above. Skinner was pretty dogmatic in thinking that the important aspects are determined by the tendency (but not universality) for people's behavior being shaped by positive and negative reinforcements.

What generates the conscious flow of aware experiences that we have? In addressing this question, what role do you see for studies of altered states of consciousness?

I'm terrified by the subject of consciousness, on both a philosophical and biological level, in that I really can't understand what people are saying most of the time. Reflecting that, I don't think consciousness is actually very important to these issues—whether you just did something with conscious intent or otherwise, it is subject to the same reality—that behavior occurred because of neurobiological events a second ago, environmental triggers a minute ago, hormone levels that morning...all the way back to your genes and why they evolved that way. This allows me to completely sidestep my dimness about consciousness.

In a 2004 article "The Frontal Cortex and the Criminal Justice System" (Sapolsky, 2004), you memorably wrote "If free will lurks in those interstices, those crawl spaces are certainly shrinking" (Sapolsky, 2004). Here we might interpret these neural "interstices" as the physical spaces surrounding different cells in the brain, and more metaphorically as the gaps in how neural systems are currently studied. In the years since that article, there have continued to be advances in functional, molecular, and systems neuroscience. How would you characterize the current state of exploration of those interstices? Over the past few years, where have we been searching where the light is, meaninglessly oversampling the same experiments or framings? And conversely, what methodologies or areas of the brain may still hold useful information?

Well, naturally, as a professorial scientist, I think that we now understand everything and nothing about the brain. All that new findings have been doing has been done to reveal more and more of the subterranean influences on our behavior—Wow, I had no idea that biology/genes/hormones/*etc.* have something to do with THAT behavior.

Where things are really problematic is that the scientific findings are on the group level, not the individual level – yes, we know with certainty that, for example, someone with their frontal cortex destroyed will behave in socially inappropriate ways—but good luck predicting which person with that damage will become a serial murderer, which will just say rude things to the host at a party. A lot of the legal scholars who weigh in with rejecting the idea that free will is a myth start at this point – if all this scientific insight can't tell me anything about what THIS individual is going to do, it has no place in the courtroom, and you sure haven't disproved the existence of free will.

My response is that we already know that, say, for every increase in the list of adversity that someone experienced in childhood (did they observe physical, sexual, or psychological abuse, did they experience it, was a family member incarcerated, was the person raised in poverty, was there substance at home… there are formal scales for quantifying this) there is about a 35% increase in the likelihood of them committing some serious (including criminal) anti-social behavior. Any time you already know enough science to say, "Someone with this background has a 93.5% chance of having been incarcerated by age 25," it doesn't matter that you can't predict events on an individual level with 100% accuracy—you don't need that to know that this is a screwed system in which people are held responsible for the uncontrollable biology that sculpted them.

In the same 2004 article (Sapolsky, 2004), you explored some of the ways in which neuroscience and philosophy intersect with justice systems, pointing to how different concepts of free will can implicitly and explicitly bear on legal proceedings. What promising and discouraging developments are you seeing at the intersection of neuroscience and law today?

Overall, I'm discouraged. I've developed a hobby of working with public defender offices to teach juries about the brain and how little control we have over who we are. I think I've worked on 11 cases, and lost ten of them. People are not open to these ideas, especially when looking at vivid pictures of corpses being shown to the jury...

Beyond the areas of law and justice: what does education look like in a world where the stance *Determined* is used as a handbook for the holistic design of learning systems?

Ugh, this is where I run into a wall. I am 100% convinced that we have no free will at all, and have thought that since I was an adolescent. But amid that, I have absolutely no idea how we are supposed to function with that insight, whether on an individual or societal level. I can truly think that way for maybe five seconds at a time. I can't visualize what things are supposed to look like if people actually understood that punishment, reward, blame, praise, any sense of someone deserving anything whatsoever, are all gibberish. I think the most important implication of that mindset, though, is that hating anyone for anything they've done is like hating an earthquake for the damage it caused, or hating a virus because it came up with a really clever spike protein and thus caused a pandemic. If we can achieve that mindset, that's some major progress.

“ I am 100% convinced that we have no free will at all (...). But amid that, I have absolutely no idea how we are supposed to function with that insight, whether on an individual or societal level. ”

Struggles with mental health, sense-making, and cognitive deterioration are becoming mainstream topics. Some of these conversations can often be distilled down to different perspectives that people take on individual choice and decision-making. What are the consequences of centering or excluding free will from these vital conversations related to individual cognition and behavior?

See above—the consequences of the two different views are monumental, and I sure can't figure out how best to run the world on a rejection of free will. The infinitely unsimple simple bit of advice is, keep that absence of free will in mind when you judge anyone about anything.

> "The infinitely unsimple simple bit of advice is, keep that absence of free will in mind when you judge anyone about anything."

After *Determined* – What are the immediate next steps, and deeper directions, for those studying free will as students or researchers? For example, are there some other topics to shift attention towards, or new philosophical tangles that can now be addressed in a better way?

This is going to keep me busy enough. Just to prepare myself, in surveys, more than 90% of philosophers firmly believe there is free will, essentially 100% of judges do, *etc.*, *etc.*, so this book is sure not going to transform much of anything.

The scientific landscape is changing for researchers in all positions, reflecting an overall turn towards collaborative approaches to education and research (Wray, 2002; Thomas & Zaytseva, 2016; Milojević *et al.*, 2018). Given your view into how career paths in science are evolving, what kind of skills or educational systems do you think will be useful for researchers in the coming years? What incentives or structures could be in place to improve career pathways for trainees in the era of Open Science (Allen & Mehler, 2019) or Decentralized Science (DeSci) (Friedman *et al.*, 2022; Hamburg, 2022; DeSci Foundation, 2021)?

My personal bias is that one of the most important things to teach, is that people realize the very limited use of reductionism in science. By reductionism, I mean basically the view that if you want to understand how something works, you can break it down into its component parts, understand how each part works, add them back together, and you will have an understanding. Instead, people need to appreciate how important chaoticism and emergent complexity are—whether at the level of neural network function, or at the level of genetic crossing at fertilization—you see that reductionism can't explain the most interesting stuff.

“ People need to appreciate how important chaoticism and emergent complexity are—whether at the level of neural network function, or at the level of genetic crossing at fertilization—you see that reductionism can't explain the most interesting stuff. ”

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# The intimacy of psychedelics, language, and consciousness.
## An interview with Jeremy I. Skipper

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## Abstract

This interview explores the intimate relationship between language and consciousness, drawing insights from aphasia phenomenology, psychedelic experiences, and neuroscientific theories. Jeremy I. Skipper, a cognitive neuroscientist, argues that language is not merely a tool for reporting conscious experiences but plays a generative role in shaping and sustaining consciousness itself. He critiques localizationist models of language processing, emphasizing the context-dependence and dynamic recruitment of brain regions. Parallels are drawn between the experiences of aphasic patients, who report a loss of self-narrative and increased connectedness, and the phenomenology of psychedelic states, which often involve a dissolution of linguistic categories and a sense of ineffability. Skipper outlines potential neural mechanisms linking language disruption to psychedelic experiences and discusses the UNITy Project, aimed in part at studying post-acute meaning-making processes and predicting changes in language and well-being after psychedelic sessions.

***Keywords**: Language, Consciousness, Psychedelics, Neuroscience, Phenomenology*

Hello Jeremy. As a specialist in the neurobiology of language, your research explores the intricate relationship between language and consciousness, and you have recently launched the UNITy

Project, short for Understanding Neuroplasticity Induced by Tryptamine, focusing on the effects of psychedelics on brain plasticity and language, among other things. I assume we will delve deeper into these topics. Could you start by telling us about your background?

I got into psychology because I wanted to figure out why I was always getting kicked out of classes. I was in a gifted program and had all of my dreams come crashing down when I was removed from it to be put into vocational school. I'm not diminishing vocational schools as I learned to weld and it was super cool. I was a really bad student and was told that I would never succeed because I couldn't think properly. It was a shock to everybody that I graduated high school.

Only in my second year of college did I become interested in consciousness. That was the cool thing that you talked about in high school and more in college, around the campfire, and with your friends, and got high and said ridiculous things about consciousness. That was what I was really jazzed about and why I went into science. I started reading obsessively. But at the end of it, no way I would get into any grad school. I didn't know if I wanted to do that. I cashed in all the bonds my grandmother gave me when I was growing up and lived in a car with a guy for two years. During that time, I read David Chalmers' (1997) The Conscious Mind and Dennett's (1993) Consciousness Explained.

At the end of two years, I knew that I didn't want to live in a car anymore and wanted to go to grad school. I was so excited about consciousness. I thought the way to get to consciousness was to get there through studying the brain. I ended up in a program, studied the brain, but ended up working on language, which isn't far removed. All of a sudden, there is like a 20-year point where I'm very narrowly focused on this very specific thing getting dumber and dumber and dumber as the years go by. Maybe there's some thread that only the gods know about. I could probably make up a story, but I think that it's almost serendipitously that I've been studying language for so long.

Recently I became unhappy with my position here and I started getting bored with what I was doing. I thought that I needed to switch gears and change careers. I even had a startup and tried different things like building sensory substitution devices for a few years, and that didn't really work out. If I was going to stay in science, I wanted to go back to where I was when I was really excited. I thus started studying consciousness again, which is the thing that I felt really compelled by around the campfire and in the car. It has just changed my life. I was not unhappy before but I was not sure I was going to stay in science and thought I really was on the verge of having some mental health problems.

Sometimes I think it's interesting to talk about mental health and how it affects lives. It is relevant only somewhat indirectly to the way I do research. I've always been into psychedelic use personally because of some of the hypomania I've had. For example, I didn't read at all until my second year of college when I started reading Tom Robbins. Through him, I learned about Buddhism and how our categories mess up the way we live. Then I started reading Alan Watts. Then I started reading obsessively at some point, probably, 5 to 10 books a week.

Switching from zero reading to doing that in a process of ingesting so much so quickly drove my interest in this idea of losing categories. Alan Watts does a nice job talking about it, especially if you're an undergrad. I grew up in a very Christian household, my mother is a lay preacher and I had very fixed ideas about what the world was until I started reading these. I didn't get this from my education. Then I got obsessive about it when I started doing psychedelics, as a way to explore those concepts. I think it's probably true of everybody so it's not really that unique.

When I hopped back into it, I started looking into the neural correlates of consciousness work. There is zero, maybe one mention and almost no discussion of the relationship between language and consciousness. Yet, philosophically there has been, and phenomenologically, we all think that there's a very strong connection somehow between language and consciousness. Dan Dennett puts it in a nice way. Actually, he's quoting somebody else who's quoting somebody else, going like: we think there's such a strong connection because we don't notice that we're conscious about things that are happening to us or within us until we verbalize it. We spend so much time doing it. I thought, "Oh my god, there's a huge hole in the literature" in the neural correlates of consciousness, which is quite comparable in the psychedelic literature.

I said: "Okay, I really love psychedelics, and I really love consciousness. I'm going to use psychedelics to perturb consciousness.". I already thought about this connection between aphasia and people's descriptions of having their self mostly disappear and having a stronger connection between them. I was maybe a little naive when I thought about it but I think when I started, I would have said there was nothing about language, the neurobiology of language, and work in the domain of psychedelics, and I wanted to use that as a tool to perturb conscious experience because of that phenomenology of oneness.

### I heard a rumor that you used to give talks barefoot. Is that true?

> "I've tried to adapt to my weirdness by becoming weirder sometimes."

That's entirely true. For 20 years in my life, I have had this weird phase in my life, starting in my second year of college where I only wore flip-flops. I even gave my job talk at UCL in a suit and flip-flops. I always wondered if I liked wearing flip-flops because it made people engage with me: "Why are you wearing flip-flops in the snow in Chicago when it's 30 below zero?" I've tried to adapt to my weirdness by becoming weirder sometimes. I literally have no answer in my own head as to why I did that, and I always make up a different story to the question "why do you wear flip-flops?". There must be something I should talk about with my therapist.

### Has that weirdness manifested in your research and work as a scientist as well?

Thanks, yes for sure. There are two clear examples which we have already spoken about. One is that from my basic science perspective, I study the neurobiology of language and 99% of the study of the neurobiology of languages is people putting people in scanners and making them listen to speech sounds or single words. I really don't think that's the only way to understand the neurobiology of language. It's probably in fact one of the worst ways.

> “I've always taken a little bit more tricky approach to the neurobiology of language. We are using it right now, it is a conversational perspective, where we actually make use of the context.”

I've always taken a little bit more tricky approach to the neurobiology of language. We are using it right now, it is a conversational perspective, where we make use of the context. From the very beginning of my career in 1999, I studied how the brain makes use of context. My first study putting people in a scanner was having them listen to videos, and movies of me telling stories about my life recorded from the head up. One of the stories is about how I used to get naked and sit in the new fountain that we got on my campus, in North Carolina during the summers. I spent quite a number of years trying to understand the visual context, if the brain makes use of gestures that people produce with their hands, mouth movements, and posture, and how we use that information to help us comprehend the words that people are saying. My approach was always to try to study language as a whole and try to reconstruct how the brain processes all of the individual putative levels of analysis like phonemes, semantics, speech production, attention, memory, etc...

What findings did you make from the perspective of context dependency of language processing?

Philosophically or ideologically, I do not think you can study language outside of the context of its use. I think it is misguided to start with the very fundamental assumption that phonemes and words have any psychological reality in the brain. I can remember this really fun meeting in Indiana with a guy named Robert Port at the end of his career. He spent his whole life studying speech sounds trying to understand phonemes: “what makes a phoneme a phoneme? How do you understand a phoneme? And how do I understand the same phoneme as a phoneme? There must be some stable acoustic properties of phonemes that allow you and me to understand the same thing when I say *dog*”. At the end of a 50-year career, he actually told me that he decided that phonemes weren't real and had no psychological reality outside of the context of reading.

> “A really interesting way of thinking about speech and language is that it is a grand delusion.”

A really interesting way of thinking about speech and language is that it is a grand illusion. Imagine hearing a language that is not a Romance language for the first time. It sounds like a continuous stream of gibberish. If you do not understand anything and there is nothing conveyed in any other channel, you have to kind of drop to the speech level. But when you think consciously about what you're hearing, you hear words, maybe sentences, and even sometimes phonemes. The reason I bring this up is that, i) it makes speech and language interesting because it is a construction in our head, and ii), it makes clear that we use different cues at different times.

I thus do not think that we have settled yet on the very fundamental components of speech and language. With the reductionist approach, stimuli need to be very tightly controlled by people pressing a button in a three-alternative forced choice task comparing phonemes such as “PA”, “KA”, “TA”. The assumption is that phonemes are such a thing that can be studied outside of the context of the rest of language and that it is going to tell you something meaningful about the neurobiology of language. I don’t think this to be true because of the ambiguity in speech sounds and a lack of invariance that is found at all levels of analysis on which we have described language. At the semantic level, words are polysemic and most words have at least five meanings. At discourse level, there's clear ambiguity in the way sentences are formed and understood. How does the brain overcome those levels of ambiguity?

I think the way that it does it is that we don't have divisibles. We don't have individual speech sounds, words, and sentences but always use language in context. We always have cues as to how a specific speech sound should be interpreted and are almost never in the dark about that. The way that we have studied speech and language has actually created this issue that maybe doesn't exist, the issue of ambiguity. In context, I can see your lips when you're talking to me which helps me decode my speech sounds, and your gestures which have semantic meaning that help me to decode what you're trying to say, I know you and have some background knowledge to help decode what you're saying, *etc.*, *etc.*, like that *ad nauseam*. To understand how people understand each other, we need to study language in the context in which we use it because context is the only place where language can be understood naturally.

I think there's room for pluralism of science here for people who study individual speech sounds or individual words, and there needs to be people who study language completely in context. I've attempted to do that at various levels of reductionism myself, but I have also done speech sound studies or with podcasts or movies where we try to construct how the brain does speech and language comprehension from the activity patterns that we see, making no assumptions about what the levels and units of analysis are. I think some of the things we found may be obvious and some of them are yet to be answered and are not obvious.

One of the obvious things that we had to establish first is that the brain uses all the context. People use gestures or mouth movements, even when they're not visible, which is something weird. Prosody is the aspect of speech that involves variations in pitch, rhythm, and intonation to convey meaning and emotional nuances. People use prosody more or less depending on prior words. When people have some background about what I am saying, people use this prior information at every given moment. We know from individual studies that the brain uses all of these and now we're starting to learn that the brain uses them all simultaneously, and more or less so depending on how informative they are. You can have an extremely informative gesture that might tell you something about people's semantic state or their emotional state and uninformative prosody. There is thus a continually changing dynamic trade-off between the cues that people use.

Your brain is always doing this dynamic dance of going up to levels and down to levels in order to be able to make use of the context that is available. One of the things we're learning now is that context is different. Context is always changing by definition, that's why we call it context. That means that the brain networks that process language are absolutely never the same.

### If that is the case, then how do we know what these networks' functions are?

> "There is currently more and more use of the term: "language network". This concept bothers me because I do not think there are stable regions in the brain."

This is going to be a little critique of fMRI and my discipline. There is currently more and more use of the term: "language network". This concept bothers me because I do not think there are stable regions in the brain. If you go back to the 19th century, from the time of Broca, who was famously associated with localizationism, we take the brain out of a deceased patient who couldn't produce speech and, by looking at his brain, we find that his inferior frontal gyrus was gone (which actually Broca's patient didn't but that's a different story). Damage to the posterior inferior frontal gyrus or the superior temporal gyrus specifically leads to problems with speech perception, language comprehension, and speech production. Yet, I think that we got it wrong for 150 years as to why that damage leads to language dysfunction.

This is even more wrong with fMRI, because fMRI is a tool based on averaging, and people use averages indiscriminately. If you're a participant in a study where you just sit there and listen to something like a news broadcast or an audiobook, you're listening and not producing because you're in a scanner. By averaging over individual words and sentences to deal with the problem of signal-to-noise ratio in fMRI, the distributed set of regions involved in speech perception and language comprehension are averaged away. All of these distributed networks that would be involved in making use of contexts, which would, by definition, be idiosyncratic because context is always changing are averaged over and you get rid of what's interesting about language, networks that vary with contexts, leaving what is stable.

What is stable tends to confirm 19th-century neurology: the posterior inferior frontal gyrus, the posterior cortex and all of the superior temporal gyrus are the home of language in the brain. What is left are regions around the auditory cortex, and connectivity hubs because 99% of the studies are auditory-based, or reading-based, so it's not surprising that you end up with regions around the auditory cortex and connectivity zones because we know the brain has a small world organization. The inferior frontal gyrus is indeed one of the biggest connectivity hubs in the entire human brain.

The inferior frontal gyrus is also a hub for other functions unrelated to language.

“It would be a mistake to call those regions the “language network” and assume that that is where language is processed only. Such claims in the literature, that we teach our students and that can be found in textbooks, are absurd.”

That is an excellent point. One of my first projects when I got to the University of Chicago was to write a list of what all of the different functions people claimed for the inferior frontal gyrus. At that time, the inferior frontal gyrus was involved in 50 or 60 different functions. People really thought that the inferior frontal gyrus was special for language, which is absolutely not the case; it is a connectivity hub for lots of things. Another hub is the Wernicke area in the posterior superior temporal cortex which is poorly defined in the language domain. Some people call it the middle temporal gyrus, or the superior temporal gyrus, or the superior temporal sulcus when it is convenient. Another connectivity hub is in the anterior temporal lobe, which is also important in language functioning, as it's part of a continuum moving out of the auditory cortex.

It doesn't mean that auditory processing regions and connectivity hubs aren't involved in language processing, because clearly if they're more active in language processing, they're doing something language-related. I just think it would be a mistake to call those regions the “language network” and assume that that is where language is processed only. Such claims in the literature, that we teach our students and that can be found in textbooks, are absurd. That is also my critique of the psychedelic discipline to some degree. I think that it is a new field and there's a lot more to be done, such as taking this naturalistic perspective that I have always had for language to apply it in the psychedelics domain.

**At the beginning of the current wave of psychedelic research when most studies were being conducted in Switzerland, neuroscience adhered to a localizationist approach. Later when we commenced our research at Imperial College, network-based approaches for fMRI were already quite well established. It is interesting how you may get sucked into speaking a language that is rebellious in one discipline but just normal in another one.**

> “We study language with fMRI, leading to the illusion that we have this specific set of regions that process language. But if we [could] study every neuron in every neuromodulatory system at once, I think we would learn that there are no language networks in the brain.”

I actually think there has even been a regression in the neurobiology of language. We were moving really in the right direction through the early 2000s into starting to think about language in the brain as distributed sets of networks that are dynamically organized. I think that has been normal for other people who think about it, after which there has been almost a regression back to thinking about it in the very localizationist sense. This is hugely problematic.

In the psychedelics field, you guys started almost out of necessity as a network-oriented field. This is not the case for the language field against which I am rebelling as we started as localizationists. Currently, we study language with fMRI, leading to the illusion that we have this specific set of regions that process language. But if we had an amazing tool where we put a human brain in a machine where we can study every neuron in every neuromodulatory system at once, I think we would learn that there are no language networks in the brain.

**What about language lateralization? Can it be tested reliably, for instance with fMRI?**

Some tests are reliable. One set of evidence comes from using the Wada test, which is an injection of an anesthetic into one of our carotid arteries, after which one hemisphere goes to sleep. Interestingly, people are more likely to go unconscious if they do the Wada test in the left hemisphere. People thought the Wada test was a gold standard for many years. My skepticism and the reason that I have a problem with the Wada test and testing laterality with fMRI is the very specific kinds of tests being used.

Wada test in the left hemisphere disturbs language more than in the right, but that depends on the kind of functions that the left and the right brain operate and goes back to why we have laterality. There has been a very long, maybe misguided, historical connection between the left hemisphere and language. I think it is misguided on many levels and depends on the test performed. People exhibit a lack of awareness that they have language problems when they obtain damage to the left hemisphere, but it's a little bit more complex than that and depends on people's language laterality.

Even amphibians have lateralized brain functions and I think that a reason for the laterality of language functioning in the human brain has to do with motor functioning. Any test that engages the motor control of speech and language is likely to be left-lateralized. But left *versus* right is too gross. There is also plenty of evidence that, for example, if you give the Wada test to the left hemisphere, people can actually still produce some words with the right hemisphere, in some cases. There are also something like 90% or more people that have left-lateralized language, even though we should really ask what that means. If we don't ask what that means, it seems to suggest that there's a really tight connection between the left hemisphere and consciousness on one hand, and the left hemisphere and language on the other hand.

The bridging assumption is that consciousness and language are related. If you have damaged, or get the Wada test, in the left hemisphere, and you have more left-lateralized language demonstrated in separate testing, you're more likely to have anosognosia of your language and more likely to go unconscious and longer to recover consciousness. If you are right-lateralized for language, the opposite is true. So if you do the right hemisphere sodium amytal injection, and you are more right-lateralized for language, you are more likely to lose consciousness, and you are slower to recover consciousness. And if you're more bilateral for language, both sides seem to be associated with consciousness, which is a really interesting set of data.

> “The idea that language is in the left hemisphere is antiquated and wrong and really depends on what you mean by language.”

The idea that language is in the left hemisphere is antiquated and wrong and depends on what you mean by language. For example, if you tested language comprehension without any motor production, language is much more bilateral. The coordination of the diaphragm and all the muscles in the lungs required to produce speech is actually the most complicated motor program that any human performs. So it is amazing that we can do it: we are all athletes. People often assume the very complex nature of language and its various levels of analysis but rarely take the complexity of the production element into account.

The more interesting aspects of language are actually, if anything, more right hemisphere lateralized: a whole set of language-related phenomena like getting jokes, making connections between things, and interestingly, the long-standing hypothesis that I do not think anybody has ever tested, that dendritic bushing and the number of connections between neurons is bigger in the right hemisphere, which is supposedly connected to the number of concepts that you can make connections with. This points towards more holistic thinking and all these hippy things that people have said for years.

### Are you planning to test those ideas of laterality with Wada tests and psychedelics?

As you may know, I do have a plan to do some self experimenting, where I would like to do the Wada test, but substitute the sodium amytal with DMT. If you go back to the split-brain studies with the first person who did the split-brain experiments (Sperry, 1962) before Gazzaniga (1998) picked it up and did a lot of the experimental work, there was this idea that goes along with the Wada test that disconnecting the two hemispheres ends up with what was initially kind of presented as a conscious left hemisphere and an unconscious right hemisphere. I think that is not the modern interpretation and some more recent views suggest that that is semi-accurate but too gross.

It is not like you end up with a conscious and unconscious brain, but you end up with two conscious brains that have different kinds of consciousness or differently distributed kinds of consciousness, which is in and of itself interesting. There are a few detractors to that idea. Recent imaging works suggest that what predicts people coming out of various levels of unconsciousness, even in a resting state, are language-related regions (Pinto et al., 2017). I review a whole ton of evidence points toward a strong connection between language and consciousness (Skipper, 2022), which may come down to how we define consciousness and left hemisphere functioning.

> “I would like to test some of my own theories by injecting DMT into the right hemisphere and maintain an observing conscious language, a self, to actually describe some of the things that have been typically called ineffable.”

I would like to test some of my theories by, just maybe for fun, injecting DMT into the right hemisphere and maintain an observing conscious language, a self, to describe some of the things that have been typically called ineffable. Imagine, in a perfect world, that you have these amazing hallucinations that arise in your right hemisphere, and you somehow still have your left hemisphere speech production capacities to describe them.

That would be amazing even though I do not think that is how it is going to work out because like I said, language is very complicated for those of us who still have our corpus callosum and rely on the right hemisphere for different kinds of functioning and taking those out could remove a lot of what's interesting about language functioning. As I said earlier, I strongly believe that a lot of why there's a strong connection between language and consciousness is the production element and that a lot of the left lateralization of language tends to be about motor functioning.

**How has the perspective on the relationship between language and consciousness evolved over time, and what role do you believe language plays in shaping our cognitive processes and consciousness?**

There was a time at various points in history when philosophers thought, like Dan Dennett, that there was a connection between language and consciousness. We have this kind of phenomenological perspective according to which there is clearly something important about language and consciousness. There's probably a reason for which I think a lot of philosophers have abandoned or semi-abandoned the idea. People often consider language just as an add-on to their cognitive systems: these modules in the brain allow you to express what's in your brain and express it to other people. It is just used with verbal reports as a tool for consciousness studies to tell us whether we're conscious or not. However, this is very misguided from the perspective of the neurobiology of language. From my perspective, I think that it is important to step back and understand the neurobiology of language, which is something I'm actually qualified to do, to make sense of the relationship between language and consciousness as compared to a perspective in which you just consider language an add-on.

To understand that, we should understand one phenomenon: you and I probably speak between 47,000 words, each, that is a lot! And, more impressively, we're using them for around 11 hours of the day! I made this calculation, I'm sure somebody will take me up on this, but I think it's actually conservative. We produce outwardly or in terms of inner speech or we're exposed to 150,000 a day, if you scale that up to a human lifespan of 70-something years, around 3,500,000,000 words. That's more input than we are exposed to in terms of faces or basically anything. From the nine months we're in the womb, we get exposed *in utero* to language that we also use *ex utero* (DeCasper & Spence, 1986). At some point before adolescence, you're picking up 150 words a day. Overall, we are exposed to a ridiculous amount of words. It would be very hard to imagine unless you have a very antiquated view of the brain, that it doesn't reshape your brain in some capacity.

There is quite a bit of research showing that even very early processes that we tend to think from textbook neuroscience to be as impenetrable from language as V1 are impacted by auditory language use, because one of its main tools, as I said earlier, is to categorize the world. For properties that are quite complex, as a dog, we ended up with a small number of categories which makes us have a very low information way to communicate with each other, which is why I think it's evolutionarily important to have language.

> “Language organizes our thoughts perceptually, emotionally, and abstractly, and is a tool for organizing the world. [...] Language is not separable from consciousness and generates and maintains conscious experience in and of itself.”

Language organizes our thoughts perceptually, emotionally, and abstractly, and is a tool for organizing the world. This is why, I think, some meditators and Buddhists ask us to give up what languages have given us, or at least just observe them. There's a little bit of tension there, I think, between meditation and psychedelics and language. Language deeply changes our brain functioning and impacts consciousness.

Then comes the question, what do you mean by consciousness? There may not be a consensus, but clearly, consciousness is not an on-and-off. Even though it seems to us that we have perceptually a unified world, consciousness is probably multifaceted, and multi-dimensional. If you do a dimensionality reduction on those multiple dimensions, you can at least say that there is something like core consciousness, which I think nobody would disagree that other animals probably have, given how similar their brains are to ours. Then you have something people might call extended consciousness, which is almost circular because it's often defined as a language you use in sort of narrative ways. We may want it to be special as humans, or something definitely human-like, and probably comes from language, from the fact that we use 150,000 words a day, and that has changed our brains.

Where my perspective may differ from other perspectives on the relationship between language and consciousness is that I consider that language is not separable from consciousness and generates and maintains conscious experience in and of itself. I think that the brain through cortico-thalamic-brainstem reverberant loops has a downward causation even on core consciousness. A very good example of this might be that by using a certain color label your whole life, words change how we perceive the visual world even without using these words (Cook et al., 2005). The low-level core process that we share with animals such as visual consciousness, if you will, visual experiences, qualia, *etc.* are clearly changed by language, meaning that language isn't just an add-on that we can call in relation to extended consciousness, it loops back onto itself, onto these kinds of earlier conceptions of conscious experiences.

**The first time we met, we talked about the phenomenology of aphasia and I just wonder if you can expand on this and how it relates to your research on language and even psychedelics.**

> “Aphasic people also experience an increased connection to everything […]. This is quite similar to the maybe more intense phenomenological aspects of psychedelics, and meditative experiences”

That is one of the things that got me into psychedelics. Anecdotally, there are probably about 10 to 20 references now of people writing about their experiences of aphasia after their recovery. Aphasia here typically means people having lost all language, inner speech, outer speech, and even some comprehension difficulties. The quite interesting phenomenological phenomenon is that they report after subsequent recovery a lack of self, at least in the narrative sense, and stop worrying about the past and the future. Language is, if nothing else, in my opinion, a tool to categorize things into objects, sensory experiences, or emotions, into labels that we use and convey to each other.

These aphasic people also experience an increased connection to everything - which seems to be more one, almost. This is quite similar to the maybe more intense phenomenological aspects of psychedelics, and meditative experiences where people lose the sense of self and experience more of a connectedness to everything. This leads to thinking about a deeper connection between language and consciousness.

Another group is feral people, these are people who were raised by ostriches and wolves, or isolated from society. Even though their experiences are anecdotal *post-hoc* reconstructions, the general consensus is that these people often lack a narrative self - the self that is comfortable fitting into social and cultural norms and that we lose with aphasia, or with psychedelics. An even deeper sort of connection to consciousness is indicated by claims that feral people fail the mirror self-recognition test.

There were mirror self-recognition tests conducted on feral people?

A lot of these tests are quite old, nothing was systematic and so far, everything we have been talking about is anecdotal and not particularly experimental; though, 10 to 20 case reports are starting to get into the realm of statistical significance. Overall, this needs to be tested and has not really been tested empirically at all as far as I know.

I assume also, children who are deaf and do not know sign language probably exist.

You are thinking of Helen Keller who was both blind and deaf and had no language input, essentially. It is one of the most eloquent descriptions of this idea that she was not conscious, as she literally said "I wasn't conscious before somebody taught me language". The problem with all these anecdotal reports is that it is hard to know what people mean by self and consciousness and self-awareness. Nevertheless, she said that she had no self until she was taught and used language, and then had thoughts for the first time. They all use those descriptions and have a similar phenomenological experience.

So then, even if you, let's say, do something that reduces your language, you're saying that your core consciousness is already modified by language? It wouldn't really, completely strip it to a pre-language state, because it's already kind of modified by language in some ways?

Yes, one hallmark of language is that we carve up the perceptual, emotional world with it. The more you use certain loops, the more they become entrenched. One cool thing is that we can use language as a tool and make poetry and songs and write novels. Those are by definition recombinations of words that have competitive, emergent meanings from them that are unique from the individual words. At a fundamental level, I have to accept that language is a causal determinant of consciousness. I was not allowed to say this in my paper (Skipper, 2022). So I can say that in an interview and be trashed for it, but I actually don't think so.

> “At a fundamental level, I have to accept that language is causal determinant of consciousness. [...] I think people are confused about what causality is, particularly in cognitive neuroscience.”

I think people are confused about what causality is, particularly in cognitive neuroscience. People often think that causality amounts to a double dissociation of necessary and sufficient conditions, and I think that's a feeble view of causality but things can be multiply caused. I do think that language can generate or cause conscious experience. If you remove language, as we've talked about, using a psychedelic, or because of aphasia, or through a particular practice (but I think I'll probably be slapped by a Zen teacher for saying that!) ... you're sitting, attending to your breath and thoughts come in.

Of course, give somebody a hammer, and everything becomes a nail; I study language, so everything conscious becomes deeply related to language because that's my hammer. They made me take their questionnaire at this aphantasia conference, and I scored the lowest possible score on this questionnaire, which makes me aphantasic. I have no visual imagery. I can remember in grad school having conversations with people at the pub like “What do you see when you close your eyes?”. I assumed that I was right from my subjective experience, that everybody else had a black experience. But I think maybe I was wrong: there is a very wide variety. Though I don't think this detracts from my own theory, actually. For example: there are aphantasic people who have claimed to have no inner speech. When I first came out with my paper, I had inner speech people saying that my theory was wrong because they don't have any inner speech, but they’re still conscious.

There's no point where I ever said that we, as animals, wouldn't be conscious if you remove language, but our consciousness would be different. It's been structured by language, our whole life. Even if you have no inner speech, you still probably have 150,000 words the day that you speak or are being spoken at. So, I always found that a weird criticism of my work. I think that consciousness is multi-determined. And I think the way that language generates and sustains consciousness isn't one thing either.

> “Going back to my own research on the neurobiology of language, I believe that words are deeply tied to every inch and every centimeter of the brain.”

I tried to elucidate the various ways in which language and inner speech generate and sustain conscious experience. There are different mechanisms and the primary one, I think, is speech production. I think that what we're primarily aware of are both inner speech and outer speech. Our words, in whatever form they come in: we're not conscious of much else that happens in our brain. We may have some imagery here and there, but I think the bulk of what a lot of people do is use a lot of language, so most of our conscious experiences revolve around language.

Going back to my own research on the neurobiology of language, I believe that words are deeply tied to every inch and every centimeter of the brain. As I described earlier, you have this set of core regions that coordinate a much more distributed set of regions. That's a very multimodal perspective. The words we hear aren't just acoustic things, they are acoustic things that are attached to the memories and the emotions associated with them. So, when you activate words, you activate a large, distributed set of regions that are not just acoustic, they activate other parts of the motor system, the sensory systems, the visual system, regions that are involved in emotional processing, *etc.* I think if that's your model of the neurobiology of language, then it has to be deeply tied to conscious experience.

### What is your understanding of inner speech and how does it relate to both consciousness and the ways we internally use language?

I'm really into inner speech as one of the ways in which we communicate with ourselves and have a kind of narrative self. There is a debate in the literature about whether inner speech is just outer speech. Inner speech is not speech production minus the actual movement of your lips and is much more complex. Inner speech varies on a continuum from engaging the motor system more or less. So, for example, we do a lot of dialogic inner speech, and we talk to ourselves about what we are doing.

> “Speech is on a continuum more or less far away from consciousness depending on being less likely to engage the motor system.”

Often we use other people's voices in our heads and communicate with those voices and have basically a dialogue in our heads in one way or another. The more it happens in sentences, the more likely it is to engage the motor system and feedback systems necessary for consciousness, whereas other kinds of speech are more forms of condensed speech, the idea that people do not speak to themselves in their head always in complete sentences. An example is soliloquy. We actually use shorthand sentences when the person we are talking to, ourselves, knows us quite well and does not really have to explain all the details to that person. Condensed forms of speech are almost our own internal language.

Speech is on a continuum more or less far away from consciousness depending on being less likely to engage the motor system. In terms of inner speech studies, people report that their inner speech is clearly using language but without the accouterments of language, without being clearly in words. It is almost like thinking in pure meaning but people do claim that phenomenologically, it is still based on language and not just imagery. This form of speech would even be further away from conscious experience because it engages the motor system less. Mind wandering is also another example of the things that typically happen at a lesser conscious level than overt speech.

### How do you see the difference between the narrative self and the images associated with the narratives?

I think the beauty of narrative is that it can be more multimodal, and, unless you mean something very different, narrative typically involves language. By connecting changes to language, at least in terms of the narrative component, these patterns then start making more sense to me as well as the phenomenology. You can have a narrative self that is you manipulating pictures of yourself in the past, but it makes zero sense to me since I don't have any visual imagery as I explained earlier. My narrative self is exclusively word-based.

> “There's still some debate and skepticism about whether there is any narrative self at all, and Strawson, I believe, claims that he has no narrative self but just episodic memories, and that's fine for me.”

There's still some debate and skepticism about whether there is any narrative self at all, and Strawson, I believe, claims that he has no narrative self but just episodic memories, and that's fine for me.

**What are the objectives of your research and how do you conceptualize the relationship between language, consciousness, and narrative self in the context of studying psychedelics?**

We have this UNITy Project: Understanding Neuroplasticity Induced by Tryptamines. My model is the HOLISTIC model (Skipper, 2022). I really wanted to move my career back to my origins to study consciousness and understand the basics of the neurobiology of the psychedelic experience acutely and then post-acutely. I thought there was an unnecessary focus on resting state data, which I now completely understand why you would collect resting state data during the acute experience.

I also wanted to use naturalistic stimuli like movies to understand the post-acute experience. I thought that something was missing from the literature, as there wasn't much of a focus on anything that happened in the brain after the psychedelic experience three years ago. The focus was more on the default mode network and interpretations around that as we've discussed. I think that should be more specific. One specificity would be what happens to language functioning during and after the psychedelic experience.

We designed the study, which is three sessions during which people come to watch a full-length movie during fMRI and do a bunch of questionnaires. In session two they take DMT in the scanner, on and off DMT during resting state. They then come back a week later, and do a second full-length movie scan of a different movie. My main interest from the very onset wasn't the acute experience, except kind of really grossly, some hypothesis around language regions being affected during the acute experience, and maybe tying those to the post-interview that we do.

My main interest was using movie scans after the psychedelic experience to understand what brain networks specifically were affected. Because I have some issues with the kind of ICA-based approaches that are typically used during resting state and I'm using a different network-based approach to parcelate the brain into much finer, detailed networks, which is one of the beauties of movie scans.

There's a lot to be done and explored there. That's what our group wants to do for the UNITy Project, at least in part. I've been linking maybe naively, from the very beginning, this aphasia phenomenology to a lesser self and a less connectedness during psychedelics. I associate language with consciousness, and particularly extended consciousness, which you might say is more about self-awareness, and meta-self-awareness, giving a deep connection to thinking about the self.

**You drew a lot of compelling parallels between aphasia and the psychedelic experience in terms of explaining the sense of connection and access to primal consciousness, via the loss of categorical thinking. How does this inform our understanding of psychedelics impact language and the sense of self?**

> “ I think nobody thinks about the self as a unitary construct anymore: there are different kinds of self. ”

I think nobody thinks about the self as a unitary construct anymore: there are different kinds of self. A gross distinction is between a bodily self and a narrative self. A lot of people in the psychedelics domain associate the narrative self with the default mode network. From a network perspective, default mode regions connect to the language regions, and a dramatic shift in default mode regions as you and others have shown should lead to a dramatic shift in the language regions. Our neuroimaging meta-analysis suggests that this may be the case.

I think that modification in language-related regions might help explain the phenomenology of psychedelics better, potentially, rather than, like whatever ego dissolution is, or thinking of the default mode network as being associated with the self or the *ego*. That's fair enough because some of the literature does outside of the psychedelics domain.

Maybe this experience of connection to everything probably comes in part from everything not being connected anymore by our language categories because categories create a separation. Neurobiologically, some of them are more or less entrenched with some neuroplasticity in the system. When you don't have those labels, you have some kind of rapid reorganization of the brain that's got to be overwhelming. Even though this is my own theory that doesn't feel like it could account for all of the experience, it must account for some of the variance in that experience, when the tool that we have for putting things into categories is diminished, if not removed.

Nobody said this to me yet, but I think, under natural psychedelic use, people probably use less language. In studies in the 50s, and 60s, they were expressly trying to understand what happens to language functioning. If you force people to use it, sometimes they talk more but their language often has more errors. It's more bizarre and unique. There are longer distance associations between words that are used, but for the most part, part of the psychedelic experience is losing language, if anything. I'm interested in what happens to language functioning during and after psychedelic use.

Your understanding of language is this kind of complex dependency. Some experiences might be completely without any relation to context, but many of them have context dependency because our brain is very context-dependent. Is there something in the psychedelic state that we cannot strip away from some of the language processes entangled in context dependency?

One of the ideas behind the experience sampling was to capture people's histories with language beforehand and see if any of that could predict their experience during and afterward. There are zero experiments that I'm aware of looking at how the exposure to language that we have, as adults day to day, through sources like Netflix, web browsing, *etc*, actually impacts our experiences down the road. There are things like monitoring, like giving babies microphones but it is hard to capture.

I wanted to do some work where we get ethical approval to just scrape everybody's internet history, their search history, therefore, all the web pages they went to, or allow us to actually track them. We actually have a student here working to see how we could predict people's subtle changes in ideology if they're reading '*War and Peace*', or '*The*

*Idiot*', which might impact them profoundly. Like for me as a 20-year-old reading Alan Watts that can probably predict a lot of my behaviors subsequent from just the words.

There's this tension here between the notion of experiences like color consciousness, or a core consciousness that can be stripped from all categories, and the argument that it is all culturally and context-dependent. According to your view, all imagery is subject to some type of story or language, even if we don't feel like we are using language, or if the language does not make any sense. From this perspective, it is inevitable, that we might be influenced by what we read about the psychedelic experience, what we read about religiosity in general, the stories we tell in life, the philosophies, etc. Based on what you're saying, psychedelics can still strip away the narrative self, but even if we just experience a core consciousness self, it has still been through a developmental kind of process through language.

Yes! You can think of the neocortex as a big memory store, to simplify it. Penfield has this paper with his theory of consciousness that was never cited. He is famous for having stimulated people's brain while they're conscious in the context of excision procedures mapping out the epicenters of epilepsy (Penfield & Jasper, 1954). When he stimulated an area in the posterior temporal lobe around the middle temporal gyrus, he gets more than words in response: a full story, for example of a woman describing herself being at a circus when a man approached her and said something (Penfield & Perot, 1963). What mattered was not only the amount of stimulation but also the connection of the region stimulated to this full memory that has been reactivated, a memory that the patients often didn't know that they knew. They recognize their own memories once they're stimulated, but they never reactivated that memory.

There is debate about how detailed these memories were but it was still pretty compelling that it was not just words but also auditory, visual, multisensory, and sometimes even olfactory associations, that you could re-stimulate at the same spot and obtain the same experience. Why do we have these detailed memories? I think it's because all of our memories and experiences are almost always associated with some form of words. So, if you, for some reason, take a psychedelic drug and you have less language function, you're still potentially going to be reactivating all of these language-associated semantics. Language "in pure meaning": somehow you have the phenomenological sense that you're using language, but it's just the meaning, without necessarily manipulating imagery either.

**Do you think that people under psychedelics experience a specific type of phenomenology, particularly in relation to the engagement of inner speech?**

Yes, when we are actually engaging in production internally, it's usually more abstract and less tied to production. The level of awareness might be associated with the extent to which the speech production system if there is such a thing is engaged. When I first decided that I wanted to use my expertise in the neurobiology of language to study consciousness using psychedelics, I went first through the literature to look at what the theoretical claims were. My cartoon view is that it was kind of centered around the default mode network. I just noted, and it was not a systematic review, that anywhere in the inferior frontal gyrus, or superior temporal gyrus, maybe posterior middle temporal gyrus, some key language cores were active. I reasoned that when we use psychedelics, we change language functioning and maybe knock out some of the auditory functioning, maybe because the visuals are actually so compelling and you spend a lot of your energy shifting the flow of blood flow to the visual system.

**So like any other immersive visual experience, the most immersive cinematic experience, your language might be less.**

It might be. If you're deeply engaged in an audiovisual task and attending to the visual, you actually get a deactivation in the auditory regions and *vice versa*, almost as if the brain has shifted its energy. My naive conception at the time is that you are probably engaged in the visual system under psychedelics, and in the language system is being either "inactive" or "deactivated", because I had no other rationale for why the language system might be affected by psychedelics, except now the consideration that maybe the distribution of 5-HT2A receptors is elevated in the temporal lobe. One of the sets of regions that were definitely affected were the inferior frontal, middle temporal, and superior temporal gyri.

You have a shift away from language-based regions, essentially a decrease in superior temporal gyrus and inferior fusiform gyrus, a phenomenology like aphasia, and a simultaneous claim in the literature about the role of the default mode network. For my domain, actually, at that time, we were talking about the default mode network in the same way as being deeply related to language functioning. As we discussed earlier, language is ambiguous, our own internal experiences need to decode the acoustic signal into the words in order to be able to understand them.

Let's talk about ineffability. There are a lot of times in a psychedelic experience that feeling of ineffability, meaning we cannot use words to describe those experiences. Is that only because these experiences are so novel and we didn't develop the language to describe them? For instance, falling in love for the first time is such an ineffable experience, like psychedelics, by just the novelty of things. They don't fit categories, and something happens to the language itself being dissolved. I wonder if you thought about this.

> "One hypothesis could be [that] vision steals the show. At the start of a DMT trip, everything in the visual experience is geometric and crazy and there's almost no room for language."

One hypothesis could be like we said earlier, vision steals the show. At the start of a DMT trip, everything in the visual experience is geometric and crazy and there's almost no room for language there. Maybe 5-MeO seems to be more cerebral and its content is a little bit more cognitive. Using different drugs may be a way to actually start carving out the role of language in these experiences. Right now, I don't feel like there's an answer. The primary hypothesis would be that the balance of forward-to-feedback connectivity is radically changing in specific regions or sets of regions such that language is being knocked out.

I think that it is clear that it is really tied to some of the experiences like the self. If you take out language, and you're not constrained by your normal categories anymore, I think that would be ineffable for us who have 150,000 words a day that we used to put people, ourselves, our feelings, what we see visually into categories. That must be a bizarre thing, something like Nagel's "What is it like to be a bat?".

My gut feeling is that something weird is happening in the language system, maybe because of the impact on 5-HT2a receptors whose distribution is not uniform and more or actually distributed throughout language core regions like the posterior inferior frontal, superior, and middle temporal gyri. I can't imagine it not being ineffable when you have these activations of the visual system and memory systems, without being constrained potentially by our normal categorical constraints.

We spoke about the effects of psychedelics on language in terms of disability, now let's mention a bit of the meaning-making because it's something so interesting to study. I'm curious actually how you understand the meaning-making process. Also, the one that happens online if there's an online meaning of making, but also after and how you're planning to study it?

For me personally, the bulk of the beauty of psychedelics is losing my whole self, the whole world becoming one and then trying to make meaning out of that based on my entire history of very specific beliefs about the world that just kind of were shattered. I really wanted to understand the post-acute process, which we've called the meaning-making process.

I thought what we could do is somehow take the movie scans and identify networks that predict changes in people's language functioning subsequent to psychedelic use, which in turn might predict changes in mental health and well-being. We're doing that by tracking people using a mobile phone-based experience sampling app, and some other methods of questionnaires on month 1, 3, 6, and 9 after the experience. We plan to first try to understand if there are any networks that predict changes in language, assuming there are any changes in language, and actually specify what those changes in language might be to try to quantify what happens to people who actually have changes who experienced positive benefits from the drugs.

We have five categories based on Hurlburt (2011) who does descriptive experience sampling with a crazy technique where he records people during the day with a very deep phenomenological analysis of people's inner thoughts. There are five different kinds of inner activity. One predictor is actually a shift away from, at least, a dialogical kind of inner speech. You might even see a gross switch between categories like between more inner speech-based thinking to a more imagistic thinking. We are thus testing a switch between types of categories of inner speech that people have described.

We know from diary and experience sampling studies that certain changes in language predict changes in mental health and well-being (Grégoire et al., 2021; Nezlek et al., 2017; Tov et al., 2013). Who is going to benefit from therapy can be predicted from changes in people's words. I think that's right now in a very nascent stage and not making use of the complexities of language. That tends to be in terms like: how many times do I refer to me? Which one might predict what changes after experiencing a change in the self, but you can go much more deeply by taking into account the balance of positive, negative, and cognitive words to predict changes in language.

Enzo Tagliazucchi had some nice ideas about using more complex metrics, like the relationship between words. If you're truly knocking out some of the categorical components of language, you might be able to see some of that reflected in the types of language people use. The study for me is an initial attempt to observe what's happening in the brain acutely and post-acutely. We can then use those observations to try to predict changes in language and quantify those in a more data-driven approach as a foundation for doing more detailed work on understanding the relationship between language and consciousness.

Currently, the experience sampling is obtained with a random beep notification five times a day on a smartphone and people speak to the phone and say exactly what they were experiencing, which may be inner speech, inner imagery, or inner feeling, which is actually an advance on previous approaches because typically, it's been text up until now. The next step to involve visual input to look for changes in micro expressions encountered some ethical issues as people were reluctant to provide their faces.

Anne Taves discusses ascription and attribution as key processes in understanding religious and psychedelic experiences (Fortier & Canna, 2018). According to her framework, ascription occurs subconsciously during the experience itself, shaped by cultural and personal contexts, while attribution is a conscious, post-experience reflection assigning specific meanings or religious interpretations. While both processes seem constructivist, they differ in their engagement with consciousness and language: ascription operates subconsciously without conscious language, yet retains a 'core self' influenced by deeper social constructs.

Given this distinction, how do these processes manifest during psychedelic experiences, particularly considering phenomena like the machine elves described by Terrance McKenna or other religious deities? Do you think these experiences are merely attributed meanings post-experience, or does the meaning-making process subconsciously shape the experience itself, without our conscious awareness?

Nick Chater (2018) has a quite controversial hypothesis called the mind is flat. Its basic premise is that we do not have the really deep unconscious minds that we think we have. He argues that what we consider unconscious processes—thought to be wild hidden forces shaping our decisions—are actually shallow, consisting merely of forgotten experiences. I don't know if this interpretation aligns with her views, but I find it difficult to accept this theory.

“ In my view, even during intense psychedelic experiences, such as those induced by high doses of DMT, where one may feel detached from reality, the language system still subtly guides the experience. ”

In my view, even during intense psychedelic experiences, such as those induced by high doses of DMT, where one may feel detached from reality, the language system still subtly guides the experience. This raises the question of whether we could hypothesize an orchestrator role for inner speech within such experiences. People on our team have developed a method for tracing experiences during our acute scans, allowing us to potentially integrate measurements of inner speech. If we analyze this data from the perspective of dynamic connectivity, rather than as static snapshots, we might observe fluctuations in the sensory-motor system responsible for speech prediction.

I expect that the absence of inner speech during psychedelic states leads to a post-acute meaning-making process, where experiences are integrated into existing categories, such as those derived from religious contexts. This aligns with my approach which emphasizes the contextual foundation of language, which earlier models often overlooked in their representations of the neurobiology of language.

All right, to finish this conversation, let's play a game where we draw from a deck of *Angel Cards* and we can both maybe speak about how those cards relate.

I start with …

… *curiosity* and *faith*.

The relationship between curiosity and faith feels deep. In my work as a researcher and in my experiences as a psychonaut, curiosity is the drive, the passion. It's also guided by this faith that things that I don't really know yet are working in some ways that we can understand.

> “I have given up on the first-person subjective experience to be a scientist and don't have much faith that we may be able to describe that first-person subjective experience.”

That could be completely misguided because your faith is founded neither on Newtonian nor Galilean specifications about what things mean in the scientific realms. I have given up on the first-person subjective experience to be a scientist and I actually don't have much faith that we may be able to describe that first-person subjective experience.

Now your turn, you have …

… *abundance* and *courage*.

I feel like I had some courage to change my career completely to study consciousness. This is scary for me, it's scary to even have this interview, or scary to talk to anybody about this. I usually refuse to talk to anybody for so long, just with you because you make me comfortable, but jumping fields to consciousness is scary and psychedelics too. I felt like I had some courage to up my career and do something new. It has led to an abundance of really cool people that I've gotten to meet, my lab, you, everybody, people who come out of the woodwork in the psychedelic domain are really interesting people who are often very kind and creative and passionate and I've been the happiest I've ever been.

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